

Matching Efficiency and Unemployment Fluctuations ^{*}

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Abstract

I document a new stylized fact about labor market fluctuations: at business-cycle frequencies, the job-finding rate is more than three times as volatile as the vacancy-filling rate. This variance ratio declines when controlling for hires from unemployment but remains far above the range predicted by standard labor search models. Moreover, the vacancy–unemployment ratio explains less than 70 percent of the job-finding rate volatility and accounts for only one-third of the average empirical slope of the Beveridge curve. These results are robust to controlling for search intensity, pointing to procyclical post-contact matching efficiency as a quantitatively important driver of unemployment fluctuations. Consistent with this mechanism, SVAR evidence shows that uncertainty shocks explain a substantial share of labor market fluctuations, with household expectations responding significantly more than firm expectations.

To reconcile these facts, I develop a labor search model with wage bargaining under incomplete information. Workers face heterogeneous expectation errors about bargaining payoffs, while firms hold rational expectations. Even with constant dispersion in workers’ expectations, incomplete information alone generates both procyclical matching efficiency and endogenous real wage rigidity, amplifying labor market fluctuations through two channels that are absent under complete information. Countercyclical dispersion further strengthens both channels. Quantitatively, the model amplifies unemployment volatility by a factor of 6.3 relative to Shimer (2005). Matching only the volatility of the vacancy–unemployment ratio, it accounts for about 80 percent of U.S. unemployment fluctuations over 2001–2025 and reproduces the unemployment–vacancy correlation and a variance ratio above one.

Keywords: Unemployment volatility, Matching efficiency, Wage bargaining, Incomplete information.

JEL Classification: D83, E24, E32, J63, J64.

1 Introduction

There is an open debate about how to account for unemployment volatility at business-cycle frequencies. The standard Diamond–Mortensen–Pissarides (DMP) framework successfully replicates the negative unemployment–vacancy relationship, but fails to account for the magnitude of unemployment fluctuations in the data (Shimer, 2005). Existing solutions mainly focus on amplifying the volatility of market tightness, defined as the vacancy-to-unemployment ($v-u$) ratio, in response to shocks (Ljungqvist & Sargent, 2017). This reflects the fact that, in the canonical DMP framework, market tightness is the only endogenous source of variation in unemployment outflows, while the passthrough from market

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tightness to unemployment outflows is governed by a stable matching technology. This technology is typically specified as a constant-returns-to-scale (CRS) matching function that maps job seekers and vacancies into hires.

However, emerging empirical evidence suggests that this mapping is not stable, and that matching efficiency—measured as the residual from the matching function—is volatile and procyclical (Barnichon & Figura, 2015; Borowczyk-Martins, Jolivet, & Postel-Vinay, 2013; Gavazza, Mongey, & Violante, 2018; Mongey & Violante, 2019; Şahin, Song, Topa, & Violante, 2014). Consistent with this evidence, I document three new facts for the U.S. labor market using data from 2001 to 2025. Together, these facts show that procyclical matching efficiency is a quantitatively important driver of unemployment fluctuations, both by directly contributing to variation in unemployment outflows and by amplifying the effect of market tightness on unemployment fluctuations.

Specifically, I first document that the joint behavior of the job-finding rate and the vacancy-filling rate in the data is systematically inconsistent with the predictions of standard labor search models with constant matching efficiency. In the data, the rank correlation between the job-finding rate and the vacancy-filling rate is far from -1 , as implied by these models. Instead, it ranges from -0.5 to -0.7 across different constructions of transition rates and margin-adjustment schemes, indicating substantial comovement between the two rates. At the same time, the job-finding rate is more than three times as volatile as the vacancy-filling rate. After controlling for hires from unemployment in the construction of the vacancy-filling rate, the variance ratio between the job-finding rate and the vacancy-filling rate is around 0.5 . By contrast, a matching function with constant matching efficiency generates a variance ratio of only 0.1 – 0.2 , depending on whether the specification is Cobb–Douglas or CES. I refer to this discrepancy as the *variance-ratio puzzle*, which complements the unemployment volatility puzzle.

Second, I find that the v - u ratio explains only 30% to 70% of the volatility in the job-finding rate at business-cycle frequencies, depending on the matching function specification and the construction of transition rates. This finding shows that variation in matching efficiency is a quantitatively important source of unemployment inflow fluctuations, beyond movements in market tightness.

Third, I show that unemployment volatility reflects not only how much the economy moves along the Beveridge curve, but also how steep the curve is. In the data, a 1% increase in vacancies is associated with a 1% decline in unemployment in the next quarter, implying a relatively flat empirical Beveridge curve at business-cycle frequencies. Decomposing this elasticity, I find that movements in the v - u ratio account for only one-third of the slope, while the v - u ratio together with separations explains roughly two-thirds. This indicates that matching efficiency plays an important role in flattening the Beveridge curve, allowing relatively small movements in vacancies or market tightness to generate large unemployment fluctuations. This also implies that the slope of the Beveridge curve is not a policy-invariant structural parameter, but depends endogenously on economic conditions that affect separation and matching efficiency.

These patterns hold after controlling for the Covid recession and the search intensity. Together, the three facts discipline the cyclical behavior of matching efficiency. While the rank correlations and job-finding rate volatility are consistent with arbitrary fluctuations in matching efficiency, jointly explaining the *variance-ratio puzzle* and the $u-v$ comovement requires matching efficiency to be procyclical. In a tight labor market, higher matching efficiency raises the probability of finding a job, so that a given change in vacancies leads to a larger change in unemployment. As a result, fluctuations in the job-finding rate are amplified and the Beveridge curve becomes flatter. At the same time, higher matching efficiency offsets congestion effects in vacancy filling and thus dampens volatility in the vacancy-filling rate. In contrast to the standard congestion mechanism, periods that are favorable for job search are not necessarily unfavorable for firms' hiring. The empirical evidence points to a systematic comovement between market tightness and matching efficiency, calling for an endogenous mechanism linking the two.

Matching efficiency operates along two margins. Conditional on the $v-u$ ratio, matching can be more efficient either because workers and firms search more intensively which increases the rate of contacts, or because contacts are more likely to result in hires. This paper focuses on the latter and defines matching efficiency as the probability of reaching wage agreement upon contact. Understanding the sources of unemployment fluctuations is important not only for advanced economies but also for emerging labor markets undergoing structural transformation.

The model extends the canonical discrete-time DMP framework with wage bargaining under time-varying incomplete information. Unemployed workers and firms with vacancies search in a decentralized labor market. The contacted worker-firm pair bargains over the lifetime wage through a modified Nash demand game: the worker and the firm simultaneously demand shares of the total net surplus from the match. If the demands are jointly feasible, the match is formed and the surplus is split at the midpoint; otherwise, no agreement is reached and the worker continues searching in the next period. The share at which agreement is reached determines the effective bargaining power of each party. Once agreed upon, bargaining power is fixed over the employment spell.

I assume that matches are homogeneous and production depends only on aggregate labor productivity. Firms observe the aggregate state and its dynamics, while workers face uncertainty about productivity and therefore hold idiosyncratic expectation errors in perceived payoffs. These expectation errors are realized before bargaining and are private information to workers. Firms hold rational expectations and have correct beliefs about the distribution of workers' expectation errors. This asymmetric information structure finds direct empirical support: using SVAR evidence on U.S. data, I show that the consumer confidence index responds roughly 10 times more strongly to uncertainty shocks than the business confidence index, and that the cross-sectional dispersion of household expectations comoves closely with aggregate uncertainty. More broadly, the cross-sectional dispersion of household expectations substantially exceeds that of professional forecasters, consistent with information frictions being concentrated on the worker side of the labor

market.

Under incomplete information, firms face a trade-off between increasing the probability of hiring and preserving expected profits per match. Greater dispersion in workers' beliefs makes it harder for firms to distinguish workers with high versus low valuations of employment, which lowers matching efficiency. To sustain hiring, firms must offer workers a larger share of the match surplus, thereby reducing firms' effective bargaining power. Consistent with the empirical evidence, I assume that dispersion in workers' beliefs is countercyclical, which implies procyclical matching efficiency and procyclical firm bargaining power, with the latter giving rise to real wage rigidity. This trade-off also operates through fundamental shocks that affect the match surplus: when the surplus is large, firms can attract workers by offering a smaller share of it, raising both the probability of agreement and firms' effective bargaining power. By contrast, in the canonical DMP model, both matching efficiency and bargaining power are constant over the business cycle.

With minimal departure from the canonical DMP framework, the model generates two additional amplification channels: procyclical matching efficiency and real wage rigidity. More importantly, uncertainty not only serves as an additional source of fluctuations, but also changes how fundamental shocks affect wage setting and matching outcomes in the labor market.

Quantitatively, I calibrate uncertainty shocks to match the volatility of the one-month-ahead uncertainty measure of Jurado, Ludvigson, and Ng (2015), and the other parameters are calibrated using Shimer (2005)'s strategy. Under this calibration, the model amplifies unemployment fluctuations by a factor of 6.3 relative to a baseline model driven only by productivity shocks. Roughly 65% of this amplification is attributable to procyclical matching efficiency, while the remaining 35% arises from real wage rigidity, which alleviates a common critique of wage-rigidity-based explanations that require implausibly high rigidity to generate sufficient unemployment volatility. When the model instead targets the observed volatility of the $v-u$ ratio, it accounts for about 80% of U.S. unemployment volatility over the period 2001–2025, a magnitude close to the contribution of the job-finding rate to unemployment volatility documented in the data (Shimer, 2012). The model also reproduces the negative correlation between unemployment and vacancies, with a correlation of -0.779 , closely matching the value observed in the data of -0.771 . In addition, it generates a variance ratio of the job-finding rate to the vacancy-filling rate of approximately 1.37, improving upon the canonical DMP models and moving the variance ratio toward its empirical counterpart.

Related literature This paper contributes to the literature by bridging recent empirical advances with the theory of unemployment fluctuations. Since Davis, Faberman, and Haltiwanger (2013), a growing literature studies how firms fill vacancies (Davis et al., 2013; Gavazza et al., 2018; Mongey & Violante, 2019), but its comovement with job finding remains unexplored. Recent work also revises estimates of the matching function to address endogeneity from procyclical matching efficiency (Barnichon & Figura, 2015; Borowczyk-Martins et al., 2013; Gavazza et al., 2018; Lange & Papageorgiou, 2020), challenging the

high matching elasticities used in many quantitative models (Hall, 2005, 2017; Hall & Milgrom, 2008; Kehoe, Lopez, Midrigan, & Pastorino, 2023; C. Pissarides, 2009). This paper studies the joint behavior of the job-finding and the vacancy-filling rates and shows that procyclical matching efficiency is a quantitatively important source of fluctuations under empirically reasonable matching elasticities.

In addition, existing theories offer limited explanations for time variation in post-contact matching efficiency. Under complete information, bargaining is efficient by the Coase Theorem and post-contact failures arise mainly from productivity mismatch, which generates little unemployment volatility (Bils, Chang, & Kim, 2011; Menzio & Shi, 2011). A notable exception is Menzio and Shi (2011), who shows that unemployment fluctuations are large when match quality is imperfectly observed at contact, highlighting the importance of informational frictions. Building on this insight, this paper introduces time-varying incomplete information into wage bargaining, generating pro-cyclical matching efficiency and real wage rigidity, and thereby contributes to understanding variations in post-contact matching efficiency.

The remainder of the paper is organized as follows. Section 2 presents the data and empirical evidence. Section 3 provides an intuitive analysis of the model mechanism. Section 4 develops a general equilibrium model of labor search with wage bargaining under incomplete information. Section 5 evaluates the quantitative performance of the model, and Section 6 concludes.

2 Empirical Evidence

2.1 Data

The data used in the analysis are quarterly averages of U.S. monthly data on key labor market variables from December 2000 to December 2025. Data on the employment (E) and unemployment level (U), flows from unemployment to employment (H_u), and short-term unemployment (U^s) are taken from the Current Population Survey (CPS). Data on the stock of vacancies (V) and total hires (H) in the nonfarm sector are taken from the Job Openings and Labor Turnover Survey (JOLTS). All series are seasonally adjusted.

The job-finding rate (JFR), denoted by f_t , is defined as the probability that an unemployed worker finds a job within period t . I consider three measures of the job-finding rate. The first measure, “CPS-Micro,” is computed as the transition probability from unemployment to employment using CPS microdata. The second measure, “EHS (2015),” adjusts the unemployment-to-employment transition probability for movements in and out of nonparticipation following Elsby, Hobijn, and Şahin (2015). The third measure, “CPS-Macro,” is constructed from the aggregate law of motion of the unemployment rate following Shimer (2012)¹, which provides a transparent link between job-finding fluctuations and unemployment dynamics. The measure “EHS (2015)” is already seasonally ad-

¹The monthly “CPS-Macro” job-finding rate is computed as $f_t = 1 - \frac{U_{t+1} - U_{t+1}^s}{U_t}$ (Shimer, 2012), where U^s is the short-term unemployment with monthly data on Number Unemployed for Less Than 5 Weeks from CPS.

justed, and the other two measures are also seasonally adjusted using X-13ARIMA-SEATS method from BLS. Figure 1 plots the time series of the different monthly measures of the job-finding rates, along with the hires-from-unemployment rate H_u/U .

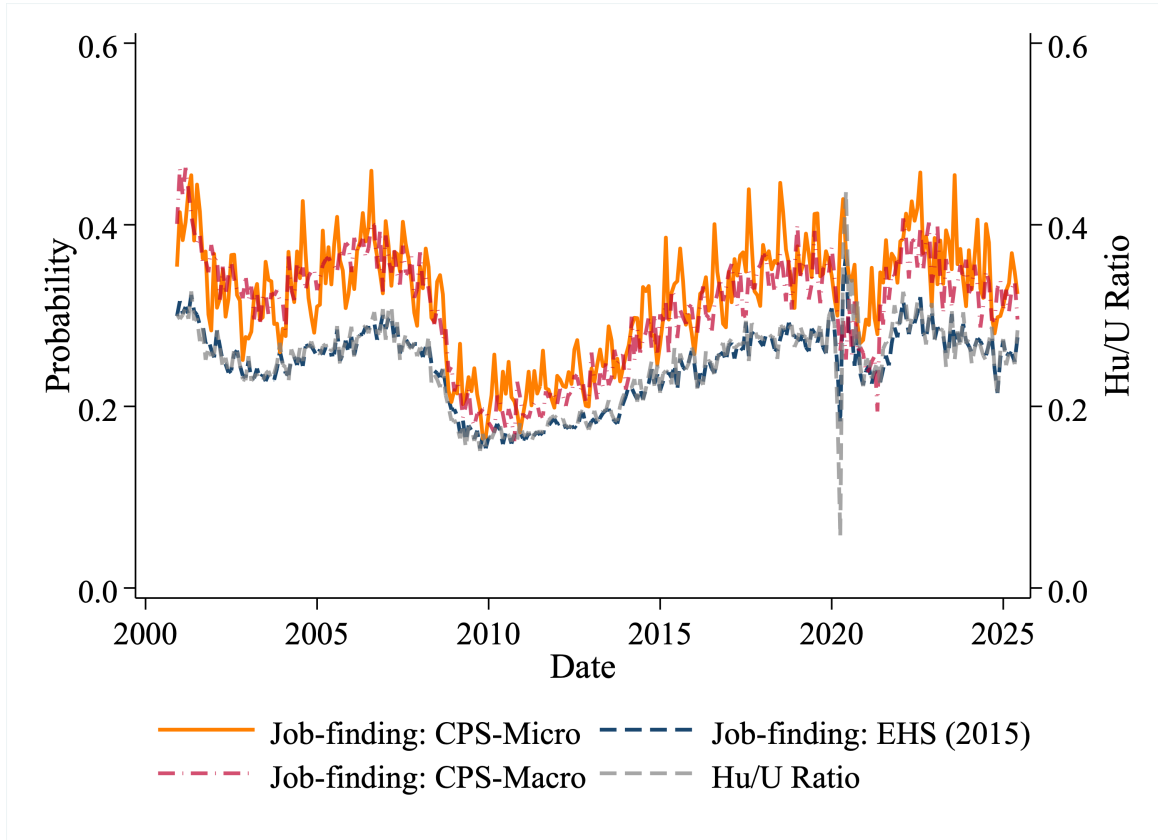


Figure 1: Comparison of different measures of job-finding rate

The vacancy-filling rate (VFR), denoted by q_t , is defined as the probability that a posted vacancy is filled during period t and is constructed following Davis et al. (2013) using aggregate data on vacancy stocks and total hires from JOLTS. For direct comparability with the job-finding rate, I also construct a vacancy-filling rate from unemployment (VFRU), denoted by q_t^u , by replacing total hires with hires from unemployment, measured using unemployment-to-employment flows from the CPS. Figure 2 plots the monthly vacancy-filling rates, along with the vacancy yield for hires from unemployment H_u/V ².

All transition rates are constructed at the monthly frequency and then aggregated by taking quarterly averages. To control for variation in search intensity, I use the aggregate measure of recruiting intensity constructed in Davis et al. (2013). To isolate business-cycle fluctuations, all variables are detrended using the Hodrick–Prescott (HP) filter with smoothing parameter $\lambda = 10^5$ (Shimer, 2005), results are robust to a Band-Pass filter.

²The vacancy yield H^u/V is not a probability measure and can exceed one because the measure of hires and vacancy stocks are from two different datasets, and because many hires are completed for vacancies created within the same month. Davis et al. (2013) document that such hires account for about 41.6% of total hires. The construction of the VFR explicitly accounts for the daily dynamics of vacancy creation and filling within the month.

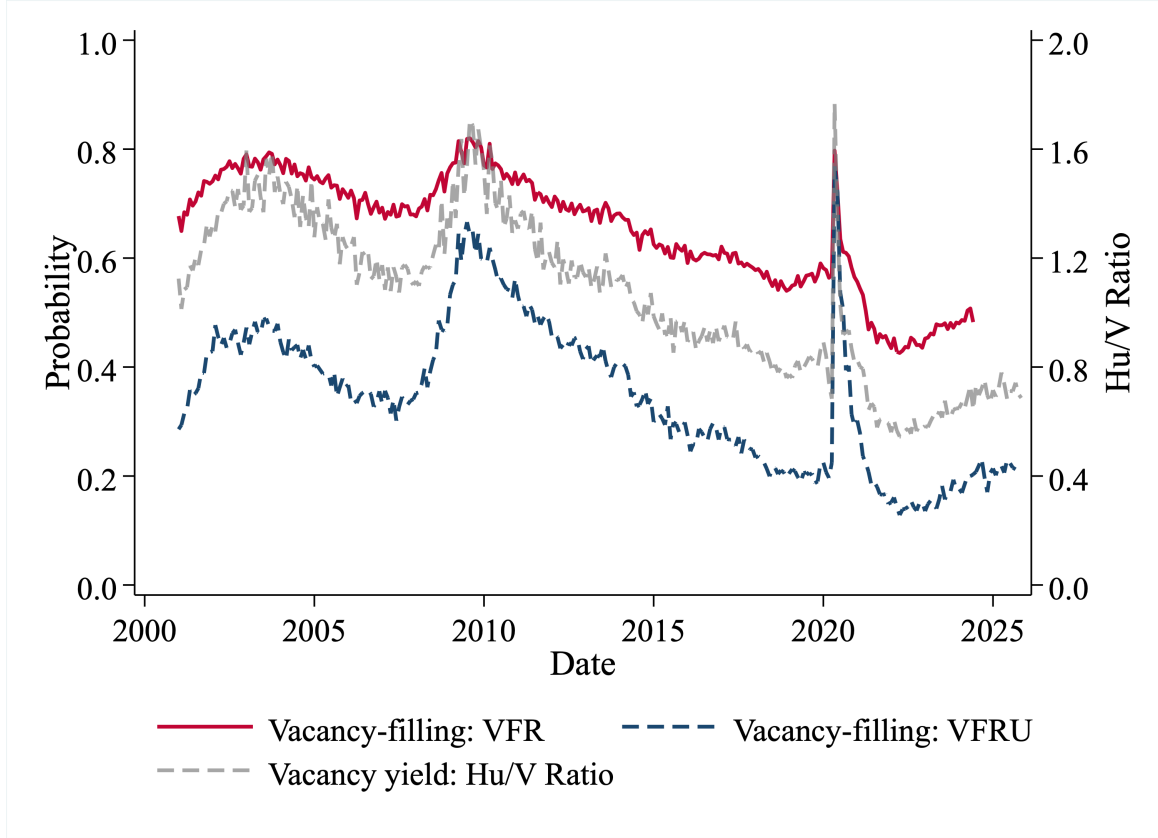


Figure 2: Comparison of different measures of vacancy-filling rate

2.2 Accounting Framework: Matching Function

To organize the empirical analysis, I adopt a standard reduced-form aggregate matching function as an accounting framework. The matching function links job finding and vacancy filling to market tightness and serves as a convenient device for decomposing labor market fluctuations into movements driven by market tightness and residual variation in matching efficiency.

Following C. A. Pissarides (1985), the process of matching job seekers to vacancies is governed by a constant returns to scale³ matching function $M(U, V)$, which is increasing in unemployment U and vacancies V . The most commonly used specification is the Cobb–Douglas matching function (Gertler, Huckfeldt, & Trigari, 2020; Hall, 2005, 2017; Hall & Milgrom, 2008; Kehoe et al., 2023; C. Pissarides, 2009; Shimer, 2005):

$$M(U, V) = \eta U^{1-\alpha} V^{\alpha},$$

where η denotes the matching efficiency, capturing variation in matching not explained by market tightness, and typically measured as the residual from the matching function. As a flexible alternative, the matching function can also be specified in constant-elasticity-of-substitution (CES) form (den Haan, Ramey, & Watson, 2000; Hagedorn & Manovskii,

³The constant-returns-to-scale assumption is standard in the literature and is broadly supported by empirical evidence (Petrangolo & Pissarides, 2001).

2008):

$$M(U, V) = \eta [(1 - \alpha)U^{\tilde{\sigma}} + \alpha V^{\tilde{\sigma}}]^{1/\tilde{\sigma}},$$

which nests the Cobb–Douglas specification as the limiting case when $\tilde{\sigma} \rightarrow 0$.⁴ Under either specification, the job-finding rate is given by

$$f = \frac{M(U, V)}{U} = M(1, \theta),$$

which is strictly increasing in market tightness $\theta \equiv V/U$, and the vacancy-filling rate is

$$q = \frac{M(U, V)}{V} = M(\theta^{-1}, 1),$$

which is strictly decreasing in θ . If the matching technology is stable over time, fluctuations in job-finding and vacancy-filling rates are driven only by movements in market tightness. In this case, tight labor markets are favorable for unemployed workers but unfavorable for firms attempting to fill vacancies because of congestion effects, which the standard view in search-and-matching models. In the empirical analysis that follows, I allow matching efficiency η to vary over time and use this framework to assess its quantitative importance.

2.3 Matching Function Estimation

Assume the matching function is Cobb-Douglas, we have the following log-linear relationship between the job-finding rate, market tightness and matching efficiency

$$\log f_t = \mu + \alpha \log \theta_t + \log \eta_t, \quad (1)$$

The OLS estimate of (1) is subject to simultaneity bias from firm's optimal vacancy creation decision (Borowczyk-Martins et al., 2013), and any expected contemporaneous change in matching efficiency will endogenously affect the v-u ratio, and thus bias the OLS estimate.⁵ To address this endogeneity, I follow Borowczyk-Martins et al. (2013) and assume that matching efficiency follows an AR(p) process⁶:

$$\log \eta_t = \sum_{i=1}^p \rho_i \log \eta_{t-i} + \varepsilon_t, \quad \varepsilon_t \sim \text{i.i.d.}(0, \sigma^2), \quad (2)$$

The shock ε_t is assumed to be i.i.d and thus uncorrelated with the lagged vacancies. Applying the lag operator $P(L) = 1 - \sum_{i=1}^p \rho_i L^i$ to both sides of (1), we have

⁴Hagedorn and Manovskii (2008) consider the special case with $\alpha = 1/2$.

⁵There is another potential bias for OLS from the non-stationarity of the variables. Dicky-Fuller test fails to reject the null of a unit root for $\log \theta_t$ and all three measures of $\log f_t$. But Augmented Engle-Granger test fails to reject the hypothesis of no cointegration. In this case, the OLS estimates may not capture the structural relationship between JFR and market tightness, but rather a spurious-regression estimator. However, Hsiao (1997) shows that the GMM estimator is consistent under maintained orthogonality conditions.

⁶I also consider the case where matching efficiency follows an ARMA process as robustness check, and the results are similar

$$\begin{aligned}
\varepsilon_t &= P(\mathbf{L}) \log \eta_t = P(\mathbf{L})(\log f_t - \mu - \alpha \log \theta_t) \\
&= \log f_t - \sum_{i=1}^p \rho_i \log f_{t-i} - \mu \left(1 - \sum_{i=1}^p \rho_i\right) - \alpha \left(\log \theta_t - \sum_{i=1}^p \rho_i \log \theta_{t-i}\right). \quad (3)
\end{aligned}$$

which implies the following moment conditions for $k \geq p + 1$:

$$\mathbb{E} [\log v_{t-k} \varepsilon_t] = 0. \quad (4)$$

where ε_t is specified as (3), and (1) can be estimated with generalized method of moments (GMM) using lagged vacancy as instruments⁷.

Similarly, the CES matching function implies following non-linear relationship

$$\log f_t = \mu + \frac{1}{\tilde{\sigma}} \log [(1 - \alpha) + \alpha \theta_t^{\tilde{\sigma}}] + \log \eta_t,$$

Applying the lag operator $P(\mathbf{L})$ to both sides, we have

$$\begin{aligned}
\varepsilon_t &= P(\mathbf{L}) \log \eta_t = P(\mathbf{L}) \left(\log f_t - \mu - \frac{1}{\tilde{\sigma}} \log [(1 - \alpha) + \alpha \theta_t^{\tilde{\sigma}}] \right) \\
&= \log f_t - \sum_{i=1}^p \rho_i \log f_{t-i} - \mu \left(1 - \sum_{i=1}^p \rho_i\right) \\
&\quad - \left(\frac{1}{\tilde{\sigma}} \log [(1 - \alpha) + \alpha \theta_t^{\tilde{\sigma}}] - \sum_{i=1}^p \frac{\rho_i}{\tilde{\sigma}} \log [(1 - \alpha) + \alpha \theta_{t-i}^{\tilde{\sigma}}] \right). \quad (5)
\end{aligned}$$

With the same moment conditions (4) and the error specification (5), the CES matching function can also be estimated with GMM. As the main purpose of this paper is not to test for different specifications of the matching function, I will focus on the Cobb-Douglas in the main text, and use CES as robustness check in the decomposition exercises. The estimation results for CES specification are reported in the Appendix⁸.

The estimation results of Cobb-Douglas matching function are reported in Table 1, where the first column reports the OLS estimates, and the rest of column reports the GMM estimates with different AR(p) specifications for matching efficiency. The estimates with different JFR measures are reported in different panels⁹.

The estimation results in Table 1 show that the OLS estimate is upward biased across

⁷As robustness check, I also consider using lagged value of job-finding rate, lagged value of unemployment rate, and the combination of these lagged variables as instruments, the results are similar, but less stable across different specifications, the results are reported in the Appendix, and the main results are based on the specification using lagged vacancies as instruments.

⁸CES nests Cobb-Douglas as a limit case with $\tilde{\sigma} \rightarrow 0$. But both NLS and GMM estimates of $\tilde{\sigma}$ is significantly negative at 1% or 5% level across specifications and JFR measures, suggesting the complementarity between unemployment and vacancies as input in matching function.

⁹See full table with AR coefficients in Appendix A

Table 1: OLS and GMM Estimates of Matching Elasticity

	OLS	AR(1)	AR(2)	AR(3)	AR(4)
Panel A: EHS(2015)					
Elasticity α	0.2251*** (0.0475)	0.2136*** (0.0084)	0.2201*** (0.0122)	0.2230*** (0.0088)	0.1792*** (0.0154)
max lag	-	1	2	-	4
$Adj. R^2$	0.6364	0.8216	0.8687	0.7319	0.8191
$Adj. R^2(\theta)$	0.6364	0.6691	0.6806	0.6865	0.6573
J-test (p -value)	-	[0.678]	[0.601]	[0.601]	[0.605]
Panel B: CPS-Micro					
Elasticity α	0.2511*** (0.0511)	0.2301*** (0.0078)	0.3237*** (0.0084)	0.3728*** (0.0121)	0.1920 (0.1631)
max lag	-	1	2	3	3
$Adj. R^2$	0.6602	0.5201	0.8186	0.7848	0.8489
$Adj. R^2(\theta)$	0.6602	0.6840	0.4921	0.3460	-9.37e+03
J-test (p -value)	-	[0.635]	[0.606]	[0.612]	[0.989]
Panel C: CPS-Macro					
Elasticity α	0.2494*** (0.0635)	0.2161*** (0.0140)	0.5654*** (0.0229)	0.1288*** (0.0187)	0.1017*** (0.0223)
max lag	-	1	2	2	2
$Adj. R^2$	0.5211	0.8262	0.6436	0.8379	0.8638
$Adj. R^2(\theta)$	0.5211	0.5519	-1.0311	0.4498	0.3835
J-test (p -value)	-	[0.593]	[0.603]	[0.591]	[0.598]
Observations	98	93	91	89	87

HAC standard errors in parentheses. OLS uses Newey–West standard errors with 12 lags. GMM is estimated by two-step GMM with HAC weighting matrix.

max lag reports the maximum lag with a statistically significant AR coefficient at the 10% level, and – means no significant lag. The full table containing AR coefficients is reported in the Appendix.

$Adj. R^2$ is computed from the standard formula based on $1 - SSR/TSS$, using the fitted value $\log \hat{f}_t = \log f_t - \hat{\varepsilon}_t$, which includes both $\log \theta_t$ and the predictable part of $\log \eta_t$. $Adj. R^2(\theta)$ is computed analogously by treating the entire $\log \eta_t$ component as residual, so that the fitted value is $\log \hat{f}(\theta_t) = \hat{\mu} + \hat{\alpha} \log \theta_t$.

J-test p -values are reported in brackets. Overidentification $df = 2$ for all specifications.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

different AR specifications. Two exceptions are with the “CPS-Micro” and “CPS-Macro” measures of job-finding rate, where the OLS estimate is higher than GMM estimates in Panel B with AR(2) and AR(3) specifications, and in Panel C with AR(2) specification. This could be because “EHS(2015)” is adjusted for the non-participation margin and composition of job-seekers, while “CPS-Micro” computed as the UE transition rate is more affected by the composition of unemployed job-seekers. During Covid, the unemployed worker on temporary layoff with large recall probability represents a large share of job seekers Gallant, Kroft, Lange, and Notowidigdo (2020). As they can relatively easy to find a job by returning to their previous employers, their job search is less elastic to the labor market tightness and generates a counter-cyclical matching efficiency, which reduces the OLS estimates and generates a downward bias compared to GMM estimates if the share of such workers is large. And “CPS-Macro” is measured by decomposing the aggregate unemployment law of motion also face the same problem. In addition, as the Covid recession is very short, estimation using quarterly data is very sensitive to the lags in AR specifications and the instruments, which could be the reason why we don’t have consistent downward bias in OLS estimate in Panel B and C of Table 1.

To control for this effect, I estimate (1) using data from 2001 to 2019, the results are reported in Table 2. After excluding Covid period, the estimates are very stable across different AR specifications and different JFR measures, see results in Table 2. As expected, both OLS and GMM estimates of matching elasticity are higher than the estimates using the full sample, and the magnitude of the upward bias in OLS estimate is also bigger. In addition, the GMM estimates are very stable across different AR specifications for matching efficiency.

This upward bias is consistent with the recent literature on procyclical matching efficiency (Barnichon & Figura, 2015; Borowczyk-Martins et al., 2013; Gavazza et al., 2018; Mongey & Violante, 2019; Şahin et al., 2014), which shows that firms endogenously create more vacancies when matching efficiency is high, and thus the OLS estimate captures both the effect of market tightness and matching efficiency.

With the full sample, the OLS estimates are around 0.22, while the GMM estimates is around 0.2 for “EHS(2015)”, but less stable for “CPS-Micro” and “CPS-Macro”, ranging from 0.15 to 0.5. But those values are still consistent with the range from 0.3 to 0.5 suggested by Petrongolo and Pissarides (2001). With sample from 2001 to 2019, the OLS estimates are around 0.3-0.35, while the GMM estimates is around 0.27-0.3, which are still around the lower bound in Petrongolo and Pissarides (2001) and consistent with recent empirical studies. Barnichon and Figura (2015) finds the OLS estimate at 0.33, and 2SLS estimate at 0.18 after controlling for matching efficiency from worker’s heterogeneity, Gavazza et al. (2018) consider heterogenous matching elasticity across subgroups of workers, and finds an average elasticity at 0.375, while the heterogenous elasticity ranges from 0.17 to 0.55. Recently, Lange and Papageorgiou (2020) uses a non-parametric approach and allows for unobserved matching efficiency, and finds a procyclical matching elasticity ranging from 0.15 to 0.3. However, the value commonly used in quantitative labor search models

Table 2: OLS and GMM Estimates of Matching Elasticity, Excluding COVID Period

	OLS	AR(1)	AR(2)	AR(3)	AR(4)
Panel A: EHS(2015)					
Elasticity α	0.3020*** (0.0468)	0.2714*** (0.0110)	0.0313 (0.1435)	0.2613*** (0.0283)	0.2748*** (0.0106)
max lag	-	1	2	3	2
$Adj. R^2$	0.7418	0.9609	0.9187	0.9468	0.9466
$Adj. R^2(\theta)$	0.7418	0.7515	0.1277	0.7739	0.7925
J-test (p -value)	-	[0.886]	[0.613]	[0.623]	[0.607]
Panel B: CPS-Micro					
Elasticity α	0.3388*** (0.0501)	0.2975*** (0.0069)	0.2890*** (0.0077)	0.2896*** (0.0069)	0.2943*** (0.0064)
max lag	-	1	2	3	1
$Adj. R^2$	0.7734	0.9533	0.9332	0.7682	0.2826
$Adj. R^2(\theta)$	0.7734	0.7809	0.7856	0.7911	0.8058
J-test (p -value)	-	[0.599]	[0.602]	[0.621]	[0.789]
Panel C: CPS-Macro					
Elasticity α	0.3544*** (0.0742)	0.2951*** (0.0147)	0.3003*** (0.0137)	0.2966*** (0.0120)	0.3103*** (0.0124)
max lag	-	1	2	2	4
$Adj. R^2$	0.6528	0.9492	0.8438	0.5344	0.6170
$Adj. R^2(\theta)$	0.6528	0.6563	0.6824	0.6974	0.7253
J-test (p -value)	-	[0.595]	[0.598]	[0.598]	[0.609]
Observations	76	71	69	67	65

HAC standard errors in parentheses. OLS uses Newey–West standard errors with 12 lags. GMM is estimated by two-step GMM with HAC weighting matrix.

max lag reports the maximum lag with a statistically significant AR coefficient at the 10% level, and – means no significant lag. The full table containing AR coefficients is reported in the Appendix.

$Adj. R^2$ is computed from the standard formula based on $1 - SSR/TSS$, using the fitted value $\log \hat{f}_t = \log f_t - \hat{\varepsilon}_t$, which includes both $\log \theta_t$ and the predictable part of $\log \eta_t$. $Adj. R^2(\theta)$ is computed analogously by treating the entire $\log \eta_t$ component as residual, so that the fitted value is $\log \hat{f}(\theta_t) = \hat{\mu} + \hat{\alpha} \log \theta_t$.

J-test p -values are reported in brackets. Overidentification $df = 2$ for all specifications.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

is 0.5 (Hall, 2017; Hall & Milgrom, 2008; Kehoe et al., 2023; C. Pissarides, 2009), suggesting that the amplification mechanisms of those models overstate the contribution of market tightness to unemployment fluctuations compared to the empirical evidence, which I will talk about more in the decomposition exercise.

In addition, the adjusted R^2 with only market tightness and the constant is around 0.5-0.65 across specifications and measures. But including the predictable component of matching efficiency increase the R^2 to around 0.8-0.85, which implies that the pro-cyclical matching efficiency explains a substantial share of variation in job-finding rate. The reason why θ -only R^2 with GMM estimates is not always lower than OLS estimate is because of the difference in sample size. After controlling for the same sample, the θ -only R^2 with GMM estimates is always lower than OLS estimates. After excluding Covid period, the θ -only adjusted R^2 is higher than the full sample, around 0.65-0.75, but the R^2 is increased to around 0.95 after controlling for the predictable component of matching efficiency, which implies that the combination of market tightness and predictable component of pro-cyclical matching efficiency explains almost all the variation in job-finding rate in the pre-Covid period.

2.4 Control for Search Intensity

Job-finding rate is the product of contact rate and acceptance rate. Aggregate market tightness determines the degree of competition on the market for each side, and explains a large share of variations in job-finding rate. But there are other factors affect the job-finding rate through the two margins separately. The contact rate is affected by how intensively workers and firms search on the market, and the acceptance rate depends on coordination between contacted worker-firm pair and how much information they know about each other. The former is well-studied in the literature, and is often referred to as “search effort” for workers and “recruiting intensity” for firms. The latter is less studied, but can also be important, as it captures the quality of matches and the likelihood of breakdowns after contact, which I refer to as the “post-contact efficiency” in this paper.

Constructing a measure of acceptance rate is challenging, as it requires detailed information on the contact process and the decision-making of both workers and firms. But there are existing measures of search intensity for both workers and firms, and after controlling for the search intensity, the remaining variation must come from the changes in post-contact efficiency. In this paper, I construct the post-contact matching efficiency as residual in matching function after controlling for search intensity.

Assume aggregate search effort of workers is s_t , and the aggregate recruiting intensity of firms is r_t . So the effective search unit is $s_t U_t$ for worker side and $r_t V_t$ for firm side. If the matching technology is Cobb-Douglas, then the number of new hires H_t is

$$H_t = \tilde{\eta}_t (s_t U_t)^{1-\alpha} (r_t V_t)^\alpha$$

and log of job-finding rate is

$$\log(f_t) = \alpha \log(\theta_t) + \alpha \log(r_t) + (1 - \alpha) \log(s_t) + \log \tilde{\eta}_t,$$

Thus, the overall matching efficiency $\log(\eta_t)$ is

$$\log(\eta_t) = \underbrace{(1 - \alpha) \log(s_t)}_{(1)\text{Search effort}} + \underbrace{\alpha \log(r_t)}_{(2)\text{Recruiting intensity}} + \underbrace{\log \tilde{\eta}_t}_{(3)\text{Post-contact}} \quad (6)$$

The equation (6) provides one way to decompose the matching efficiency into three margins: (1) worker's search effort, (2) firm's recruiting intensity, (3) unobserved post-contact matching efficiency $\log \tilde{\eta}_t$. Given the measure of (s_t, r_t) , I could back out (3) as the residual:

$$\log(\hat{\tilde{\eta}}_t) = \log(\hat{\eta}_t) - \hat{\alpha} \log(r_t) - (1 - \hat{\alpha}) \log(s_t)$$

The log-linearity of (6) allows variance decomposition of total matching efficiency into the three margins, and for each margin x_t , the variance share in total matching efficiency is simply $\frac{\text{Cov}(x_t, \log \eta_t)}{\text{Var}(\log \eta_t)}$. However, for the CES matching function, such linearity is absent and we can only disentangle post-contact margin from the pre-contact margin. Under CES matching function, the log job-finding rate is

$$\log(f_t) = \frac{1}{\hat{\sigma}} \log[(1 - \alpha) + \alpha(\theta_t^e)^{\hat{\sigma}}] + \log(s_t) + \log(\tilde{\eta}_t)$$

where $\theta_t^e = (r_t V_t)/(s_t U_t)$ is the effective market tightness. Therefore, the post-contact efficiency $\tilde{\eta}_t$ can be computed as the residual

$$\log(\hat{\tilde{\eta}}_t) = \log(\hat{\eta}_t) - \underbrace{\left[\frac{1}{\hat{\sigma}} \log \left[\frac{(1 - \hat{\alpha}) + \hat{\alpha}(\theta_t^e)^{\hat{\sigma}}}{(1 - \hat{\alpha}) + \hat{\alpha}(\theta_t)^{\hat{\sigma}}} \right] + \log(s_t) \right]}_{\text{Pre-contact margin}}$$

The literature finds that recruiting intensity r_t is procyclical, while the search effort s_t is counter-cyclical, but the cyclicity of post-contact efficiency $\tilde{\eta}$ and its contribution in business cycle fluctuations remains unknown. To the best of my knowledge, this paper is the first to disentangle these three margins and quantify their relative contributions to business-cycle fluctuations in matching efficiency.

For aggregate search effort s_t , I follow Mukoyama, Patterson, and Şahin (2018), and construct s_t from the average search minutes per day of unemployed worker. I first estimate the relation between search methods weeks and search time using ATUS data from 2003-2024, then impute the search time for CPS sample from 1994-2024 using the estimates in first step.¹⁰ I then normalize the average search time by the sample mean of 21 minutes

¹⁰The original series and first-step estimates in Mukoyama et al. (2018) is only available until 2014, but the imputed aggregate search effort with first-step estimates using ATUS 2003-2014 sample is very close to the one using ATUS 2003-2024 sample. The correlation for level data is 0.937, and the correlation for HP cyclical component is 0.984, which implies that the structural relation between search time and method is very stable over time, and one can directly adopt their 2014 estimates to construct the aggregate search effort from CPS

such that on average the effective search unit is 1 for unemployed worker.

For the recruiting intensity, I use the measure from Davis et al. (2013) and Mongey and Violante (2019). Both are model-based series. Davis et al. (2013) estimates the elasticity of firm's vacancy-filling rate to recruiting intensity with establishment level data, and assume the macro elasticity is the same as micro elasticity to compute the aggregate recruiting intensity. Mongey and Violante (2019) instead derive the recruiting intensity from firm's optimal decision problem. In both papers, the matching function is in the following form

$$H_t = \Phi_t A_t V_t^\alpha U_t^{1-\alpha}$$

where Φ_t is the aggregate recruiting intensity, and A_t capture the other margins in matching efficiency. So their measure Φ_t is essentially r_t^α in (6)¹¹. As both relies on the specific model assumptions and have comparative advantage, I consider both in the main analysis to see whether the decomposition is sensitive to different construction methodology.

The variance decomposition for each JFR measure, recruiting intensity measure and sample is reported in Figure 3, and the sum of the variance share is equal to 1 by construction¹². Due to the Covid effect, the share of counter-cyclical worker's search effort takes a large share in total variance of matching efficiency, after excluding Covid period, the variance share of search effort is stable and around 20%-25% across JFR and recruiting intensity measures. Firm's recruiting intensity is an important margin in pro-cyclical matching efficiency, and its contribution is around 40%-50% in total variance. However, as r_t and s_t partially offset each other, the post-matching efficiency margin is quantitatively important and explain at least 70% of variations in total matching efficiency.

2.5 Decomposition Exercises

The pro-cyclical matching efficiency has three important implications: (1) OLS estimates overstate the variance contribution of market tightness to job-finding rate and thus to unemployment volatility, (2) it does not only shifts the Beveridge curve but also flattens the curve as unemployment outflow changes more in response to in vacancy creation, (3) it affects the comovement between JFR and VFR in a way that amplifies volatility in JFR and dampens volatility in VFR as the spillover effects for firms in a tight labor market is partially offset by high matching efficiency.

Using the matching function as a decomposition device, I will quantify the effect and contribution of matching efficiency on these aggregate moments. The results suggest that the fluctuations in pro-cyclical matching efficiency is a quantitatively important component of business cycle, and it contributes the by amplifying the volatility of JFR.

data.

¹¹Mongey and Violante (2019) consider different values of α to construct Φ_t , I choose the one with $\alpha = 0.25$ which is close to the empirical estimates I have in Table 1 and 2. In addition, they also provides an aggregate index Φ_t^{index} for Φ_t , which depends only on total hires and total employment and thus allows me to extend their series to 2025.

¹²Variance share across all estimation methods is in Appendix. The results are similar.

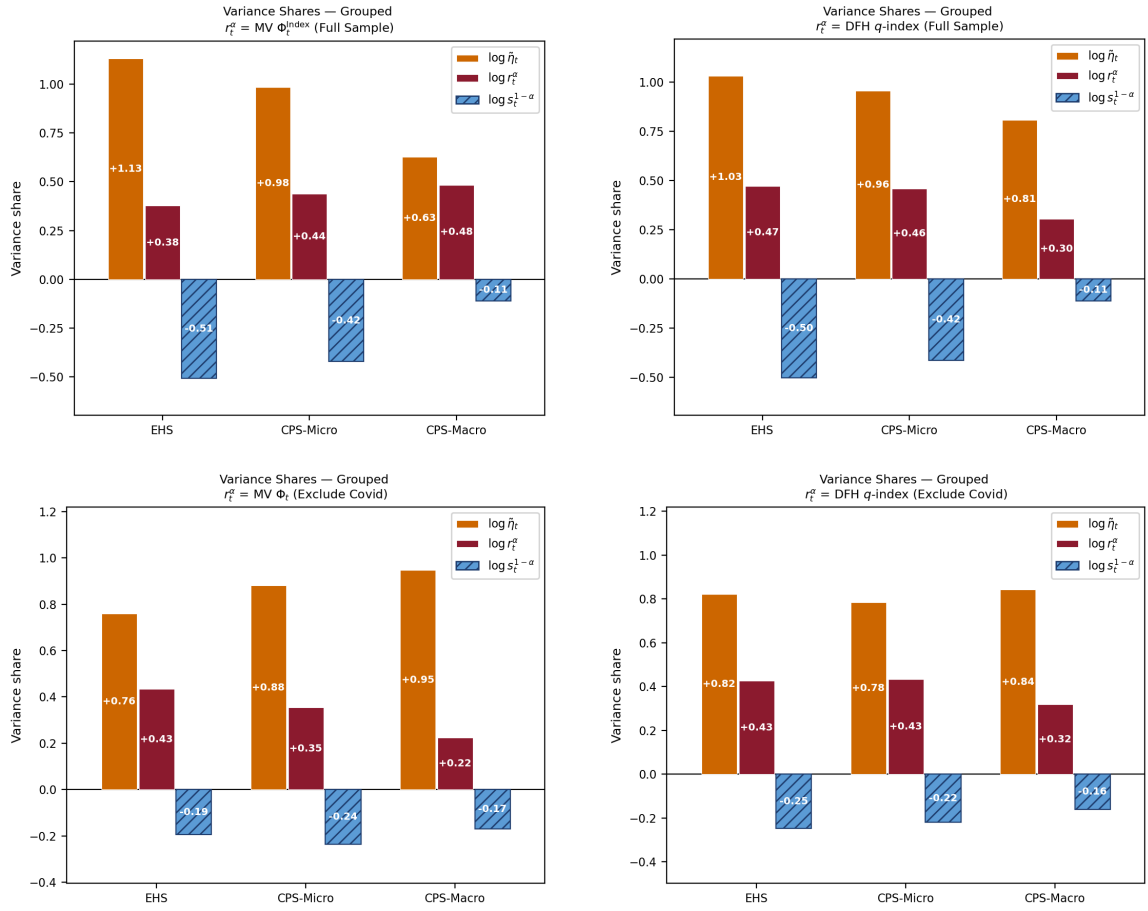


Figure 3: Variance shares of the three margins of matching efficiency.

2.5.1 Fact 1: Decomposition of Variance of JFR and Unemployment

The first question is how much of the volatility in JFR and unemployment is explained by fluctuations in matching efficiency at business cycle frequencies. To answer this question, I first construct series of counterfactual JFR and unemployment from the fitted value of GMM estimation, then compute the variance ratio of the business cycle component of the observed data and the counterfactual series. I consider both Cobb-Douglas and CES matching function with GMM estimates, and use HP filter with smooth parameter at $\lambda = 10^5$ (Shimer, 2005) to, as well as Band-Pass filter targeting 6-32 quarter in the frequency domain as a robustness check.

For example, if matching function is Cobb-Douglas, I use the fitted value $\hat{f}_t = \exp(\hat{\mu}_{GMM} + \hat{\alpha}_{GMM} \log \theta_t)$ as the counterfactual JFR, and consider two counterfactual unemployment series based on the law of motion of unemployment

$$U_{t+1} = (1 - f_t)U_t + x_t E_t, \quad t = 0, \dots, T,$$

where f_t and x_t are the discrete-time job-finding and separation rate, E_t is the level of employment. The two counterfactual series are

1. θ -only: $\hat{U}_{t+1}^\theta = (1 - f_t)\hat{U}_t^\theta + \bar{x}E_t$, for $t = 1, \dots, T$.
2. θ + Separation: $\hat{U}_{t+1}^{\theta,x} = (1 - f_t)\hat{U}_t^{\theta,x} + x_t E_t$, for $t = 1, \dots, T$.

where $\hat{U}_0^\theta = \hat{U}_0^{\theta,x} = U_0$, $\bar{x}$ is the average separation rate over the sample period. In the first counterfactual, $\{\hat{U}_t^\theta\}$, the only variation in unemployment outflow is changes in market tightness, and the only variation in unemployment inflow is changes in employment; in the second counter-factual, $\{\hat{U}_t^{\theta,x}\}$, both market tightness and separation rate varies as in the data, and only matching efficiency is kept constant. This decomposition helps to discipline the three margins (θ_t, η_t, x_t) , and evaluate their relative contribution to fluctuations at business cycle frequencies. The gap between the variance of actual series and “ θ +Separation” counterfactual is the part explained by matching efficiency.

For the GMM estimates, I choose the specifications with largest significant lags for the margin-adjusted measure “EHS(2015)” as main specifications. The results with HP filter are reported in Table 3, and the results with Band-Pass Filter and other estimates are similar and reported in Appendix A.

Under Cobb-Douglas matching function, market tightness alone can only generate around 30% of JFR volatility, this share is increased to 47% for pre-Covid period. In addition, the fluctuations in market tightness also account only for 23%-25% of unemployment volatility, while for 45%-50% for the pre-Covid period. Augmenting with actual separation rate, the matching efficiency alone explains around 34%-46% fraction of unemployment volatility in the full sample, and around 20%-37% of unemployment volatility for pre-Covid period. This exercises shows that fluctuations in pro-cyclical matching efficiency is a quantitative important component for business cycle volatility. With CES matching function, the variance share of matching efficiency in both JFR and unemployment rate

Table 3: Variance Shares: Role of Market Tightness and Separation, HP Filter

	EHS (2015)	CPS-Micro	CPS-Macro
<i>Panel A: CD, Full Sample (GMM AR(4))</i>			
JFR: θ -only	0.2881	0.2640	0.0576
U : θ -only	0.2556	0.2611	0.2370
U : θ + Separation	0.6434	0.6624	0.5421
<i>Panel B: CD, Exclude Covid (GMM AR(3))</i>			
JFR: θ -only	0.4749	0.4765	0.4093
U : θ -only	0.4526	0.4821	0.4902
U : θ + Separation	0.7337	0.7919	0.8071
<i>Panel C: CES, Full Sample (JFR: AR(3), U: AR(4))</i>			
JFR: θ -only	0.7824	0.6307	0.5428
U : θ -only	0.2260	0.1418	0.3016
U : θ + Separation	0.5745	0.4800	0.7298
<i>Panel D: CES, Exclude Covid (GMM AR(2))</i>			
JFR: θ -only	0.3733	0.4209	0.3486
U : θ -only	0.0938	0.1228	0.1293
U : θ + Separation	0.3357	0.4051	0.4143

Notes: All entries are variance shares $(SD_{cf}/SD_{actual})^2$, HP filter with $\lambda = 100,000$. JFR: $\text{Var}(\hat{f}_{\theta, cyc})/\text{Var}(f_{cyc})$, share of job-finding rate variance explained by market tightness (theta-only fitted value). U_{θ} : $(SD(\ln u_{cyc, \theta}^{cf})/SD(\ln u_{cyc}))^2$, share of unemployment variance explained by tightness-driven JFR with mean separation (CF-fixX). $U^{\theta, x}$: $(SD(\ln u_{cyc, \theta, x}^{cf})/SD(\ln u_{cyc}))^2$, same but with actual separation rate (CF-theta). JFR measures: EHS (2015) = `lnpue`, CPS-Micro = `lnjf_sa`, CPS-Macro = `lnf_sa`.

significantly improved, but there is still around at least 20% of JFR volatility and at least 30% of unemployment volatility is explained by the residual matching efficiency. Similarly, for the pre-Covid period, the variance share of market tightness to JFR is similar to the results with Cobb-Douglas matching function, and even smaller, the resulting variance share in unemployment is also even smaller.

The variance shares of effective market tightness $\theta_t^e = (r_t V_t)/(s_t U_t)$ in JFR and unemployment are reported in Table 4. Augmenting with search intensity substantially increase the volatility of counterfactual JFR, and thus the explained share in unemployment also increased by 10 to 20 percentage point, but effective market tightness and separation alone still cannot fully explain the unemployment volatility.

Table 4: Variance Shares: Role of Search Intensity and Recruiting Intensity, HP Filter

	EHS (2015)	CPS-Micro	CPS-Macro
<i>Panel A: DFH q-index, Full Sample (GMM AR(4))</i>			
JFR: θ^e -only	0.9341	0.8030	0.3837
JFR: $\tilde{\eta}$ -only	0.8023	0.6457	0.5886
U : θ^e -only	0.3373	0.3447	0.3085
U : $\tilde{\eta}$ -only	0.1108	0.1655	0.1594
U : θ^e -only + Sep	0.8621	0.8796	0.7684
U : $\tilde{\eta}$ -only + Sep	0.1242	0.1796	0.3356
<i>Panel B: DFH q-index, Exclude Covid (GMM AR(3))</i>			
JFR: θ^e -only	0.8461	0.8118	0.6904
JFR: $\tilde{\eta}$ -only	0.1333	0.1043	0.1124
U : θ^e -only	0.6060	0.6484	0.6598
U : $\tilde{\eta}$ -only	0.0204	0.0351	0.0618
U : θ^e -only + Sep	0.9724	1.0447	1.0636
U : $\tilde{\eta}$ -only + Sep	0.0654	0.0810	0.1261

Notes: All entries are variance shares $(SD_{cf}/SD_{actual})^2$, HP filter with $\lambda = 100,000$. Fitted JFR = $\log f_t - \log \tilde{\eta}_t$, where $\log \tilde{\eta}_t$ is the residual from the three-margin decomposition $\log \eta_t = (1 - \alpha) \log s_t + \log r_t^\alpha + \log \tilde{\eta}_t$. Panel A: DFH q-index, Full Sample α , GMM AR(4). Panel B: DFH q-index, Exclude Covid α , GMM AR(3), sample 2003Q2–2019Q4. JFR θ^e -only: $(SD(\hat{f}_{cyc}^{\theta^e})/SD(f_{cyc}))^2$; fitted JFR = $\log f_t - \log \tilde{\eta}_t$ (removes $\tilde{\eta}_t$, retains θ, s, r). JFR $\tilde{\eta}$ -only: $(SD(\hat{f}_{cyc}^{\tilde{\eta}})/SD(f_{cyc}))^2$; fitted JFR = $\log f - \log \tilde{\eta}_t + \log \tilde{\eta}_t$ (only $\tilde{\eta}_t$ varies, θ, s, r fixed at means). U rows: $(SD(\ln u^{cf})/SD(\ln u))^2$; CF-fixX uses mean separation, +Sep uses actual separation (CF- θ). JFR measures: EHS (2015) = `lnpue`, CPS-Micro = `lnjfsa`, CPS-Macro = `lnf_sa`.

2.5.2 Fact 2: Decomposition of Slope of Beveridge Curve

The next question is how does the matching efficiency shape the Beveridge curve and thus the equilibrium u-v relation. The answer to this question provides onw way to disentangle the potential amplification mechanism for unemployment volatility. A lot of work has been done in investigate the relation between the shifts in Beveridge curve and structural change in matching efficiency. But if the a nontrivial part of change in matching efficiency occurs at higher frequency, e.g., business cycle frequencies, it will also affects the slope of the Beveridge curve. The GMM estimates suggests the measured matching

efficiency is pro-cyclical which amplifies the increase in JFR and thus the drop in unemployment when there are more vacancies on the market. Therefore, the Beveridge curve should also be flatter, and the equilibrium elasticity of unemployment to vacancy is greater in magnitude.

Based on this intuition, I computed the observed elasticity of different series of unemployment U_t to lagged vacancies V_{t-1} , both detrended with HP filter with smooth parameter $\lambda = 10^5$. This elasticity is an estimate of average slope of cyclical Beveridge curve, and has no causal interpretation. For the unemployment series, I use the actual unemployment $\{U_t\}$, the “ θ -only” counterfactual unemployment $\{U_t^\theta\}$, and the “ θ +Separation” counterfactual unemployment $\{U_t^{\theta,x}\}$.

Table 5: Beveridge Elasticity $\partial \ln u / \partial \ln v$: Summary, HP Filter

	EHS (2015)	CPS-Micro	CPS-Macro
<i>Panel A: CD, Full Sample (GMM AR(4))</i>			
U_t (actual)	−0.9764*** (0.1550)		
U_t^θ	−0.3084*** (0.0399)	−0.3350*** (0.0416)	−0.1535*** (0.0362)
$U_t^{\theta,x}$	−0.6672*** (0.0913)	−0.7000*** (0.0963)	−0.5054*** (0.0845)
<i>Panel B: CD, Exclude Covid (GMM AR(3))</i>			
U_t (actual)	−1.1265*** (0.1185)		
U_t^θ	−0.5016*** (0.0745)	−0.5622*** (0.0800)	−0.5772*** (0.0814)
$U_t^{\theta,x}$	−0.8708*** (0.0885)	−0.9334*** (0.0907)	−0.9489*** (0.0913)
<i>Panel C: CES, Full Sample (GMM AR(3))</i>			
U_t^θ	−0.3176*** (0.0793)	−0.3750*** (0.0786)	−0.3935*** (0.1001)
$U_t^{\theta,x}$	−0.5879*** (0.1489)	−0.6945*** (0.1434)	−0.7189*** (0.1615)
<i>Panel D: CES, Exclude Covid (GMM AR(2))</i>			
U_t^θ	−0.2497*** (0.0860)	−0.3789*** (0.0897)	−0.3956*** (0.0912)
$U_t^{\theta,x}$	−0.5777*** (0.1278)	−0.7334*** (0.1049)	−0.7548*** (0.0993)

Notes: Each cell reports the coefficient $\hat{\beta}$ from $\ln u_t^{\text{cyc}} = \beta \cdot \ln v_{t-1}^{\text{cyc}} + \varepsilon_t$, with moving block bootstrap standard errors in parentheses ($B = 8$ quarters, 499 replications). HP filter with $\lambda = 100,000$. U_t^θ : counterfactual unemployment with theta-only JFR and mean separation (CF-fixX). $U_t^{\theta,x}$: counterfactual unemployment with theta-only JFR and actual separation (CF-theta). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The results are reported in Table 5. In the data, 1% increase in vacancy is followed by an increase of roughly 1% in unemployment in the next quarter. This elasticity is slightly

higher for pre-Covid period. Matching efficiency explains only 1/3 of this elasticity in the full sample, and less than 1/2 in the pre-Covid period. Market tightness and separation together explains around 2/3 of this elasticity in the full sample, and around 77%-85% of this elasticity in the pre-Covid period. It is unambiguous that constant matching efficiency results in a steeper Beveridge curve, and smaller volatility in unemployment at business cycle frequencies. And the pro-cyclical matching efficiency explains around 1/3 of the slope of Beveridge curve from 2001 to 2025, and around 1/6 for the pre-Covid period. Even though CES increase the variance share of market tightness to JFR, it does not significantly improve the imputed u-v elasticity. Therefore, allowing a more flexible specification of matching function does not help to explain the flatness of Beveridge curve.

This decomposition suggests that to explain the observed unemployment volatility, we need a model that is able to generate both flat enough Beveridge curve and also enough fluctuations along this Beveridge curve, which indeed from a missing component, i.e. the pro-cyclical matching efficiency, rather than from the choice of matching function specification. The theoretical literature contributes a lot on the second ingredient, but they rather assumes this slope as a structure parameter and relies on an ad-hoc high matching elasticity $\alpha = 0.5$. This paper is trying to complementing the existing literature by providing micro-foundation for the flat Beveridge curve.

The estimated u-v elasticity with effective search intensity for search intensity is reported in Table ???. After controlling for search intensity, the estimate u-v elasticity increased in magnitude by 0.1-0.2, but still below the actual elasticity.

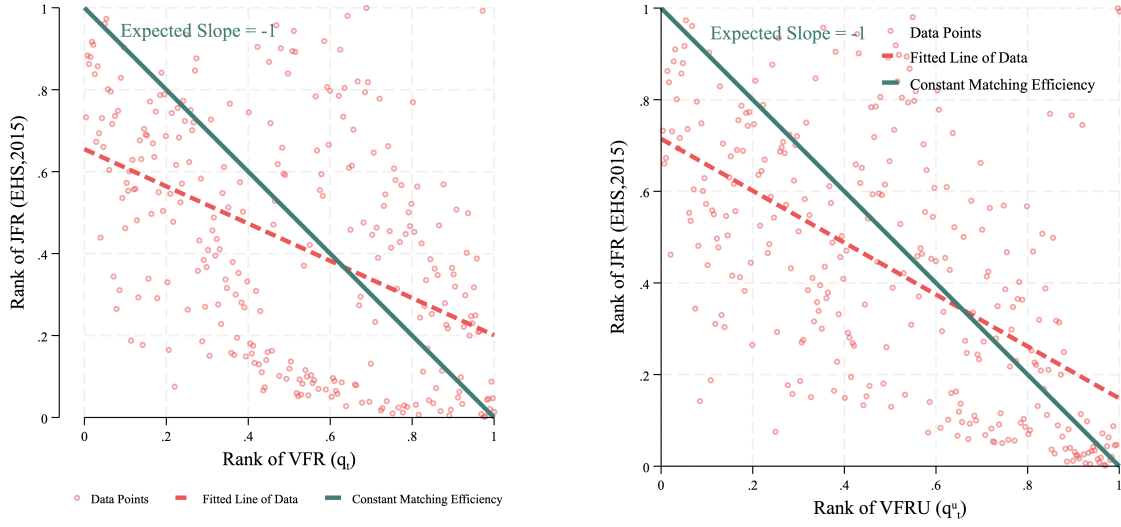
2.5.3 Fact 3: Comovement between JFR and VFR

The third question is how the matching efficiency affects the comovement between JFR and VFR. Standard labor theory predicts that market tightness is the dominant driving force of matching process, and a tight labor market is good for worker's job search but bad for firm's recruiting because there is more competition on the firm side. However, matching efficiency affects both JFR and VFR in the same direction, thus generates positive correlation between the two.

Figure 4 provides a visual summary of the comovement between job finding and vacancy filling. Panel A plots the rank of "EHS(2015)" JFR against the rank of the vacancy-filling rate (VFR). Panel B replaces VFR with the vacancy-filling rate based on hires from unemployment (VFRU). The figure shows that the relationship is far from a one-for-one negative mapping: even in ranks, job finding and vacancy filling exhibit substantial dispersion around any monotone decreasing relationship. Figures constructed using CPS-Micro and CPS-Macro measures display a similar pattern and, if anything, exhibit even larger deviations from a perfect negative rank correlation.

To quantify the comovement between job finding and vacancy filling, I compute both the Spearman rank correlation between the log job-finding rate and the log vacancy-filling rate, and the Pearson correlation of their business-cycle components obtained using the HP filter. The results are reported in Table 6.

Figure 4: Rank comovement between job finding and vacancy filling



(a) EHS(2015) JFR vs. VFR

(b) EHS(2015) JFR vs. VFRU

Notes: Each panel plots the rank of the EHS(2015) job-finding rate against the rank of the vacancy-filling rate measure. Panel A uses the baseline vacancy-filling rate (VFR). Panel B uses the vacancy-filling rate based on hires from unemployment (VFRU). Ranks are computed for monthly data from January 2001 to September 2005.

Table 6: Comovement between job-finding rates and vacancy-filling rates

	Job-finding rate (JFR)		
	(1) CPS-Micro	(2) EHS (2015)	(3) CPS-Macro
<i>Panel A: Spearman rank correlation</i>			
VFR ($\ln q_t$)	-0.472	-0.504	-0.282
VFRU ($\ln q_t^u$)	-0.619	-0.643	-0.518
<i>Panel B: Correlation of cyclical components</i>			
VFR ($\ln q_t$)	-0.552	-0.579	-0.658
VFRU ($\ln q_t^u$)	-0.645	-0.680	-0.736

Notes: Each entry reports the correlation between a vacancy-filling rate measure (row) and a the job-finding rate measure (column). Panel A reports Spearman rank correlations between the log level of JFR and VFR, Panel B reports Pearson correlations of their cyclical components, detrended by HP filter with smooth parameter 10^5 . The sample has 99 quarterly observations from 2001Q1 to 2025Q3.

Under constant matching efficiency, the job-finding rate and the vacancy-filling rate should exhibit a perfect negative rank correlation of -1 . In the data, this is clearly not the case. Across alternative measures, the rank correlation is much weaker, typically between -0.3 and -0.5 . When vacancy filling is restricted to hires from unemployment, the correlation becomes more negative, falling in the range of -0.5 to -0.65 . This reflects the strongly countercyclical nature of hires from unemployment, which mechanically strengthens the negative comovement. Nevertheless, even under this restriction, the correlation remains far from the value implied by a stable matching function. Correlations of the cyclical components shows a similar pattern.

Beyond comovement, the data also reveal an asymmetry in the volatility of job finding and vacancy filling. The first two rows in each panel Table ?? reports the standard deviations and variance ratios of the cyclical components of different measures of job-finding and vacancy-filling rates observed in the data. Across all measures, the cyclical component of the job-finding rate is substantially more volatile than that of the vacancy-filling rate. The resulting variance ratio ranges from 4.2 to 6.8, depending on the measure used, and this variance ratio is even larger for the pre-Covid period, which is consistent with the evidence on surge in recalls and thus countercyclical matching efficiency during Covid. When vacancy filling is restricted to hires from unemployment, the vacancy-filling rate becomes more volatile, reflecting the large cyclical fluctuations in hires from unemployment, and the variance ratio declines to a range between 0.3 and 0.5 for the full sample, and between 0.6 to 0.9 for pre-Covid period. This qualitative pattern is robust to alternative filtering methods and other estimates of matching function.

Table 7: Variance Ratio $(SD_f/SD_q)^2$: Summary, HP Filter

	EHS (2015)	CPS-Micro	CPS-Macro
<i>Panel A: Full Sample (GMM AR(4))</i>			
Data: f_t vs q_t (total hires)	4.2286	5.4193	6.8729
Data: f_t vs q_t^u (UE hires)	0.3350	0.4293	0.5445
CD: $\hat{f}_t^\theta$ vs $\hat{q}_t^\theta$	0.0477	0.0564	0.0128
CD: $\hat{f}_t^{\theta^e}$ vs $\hat{q}_t^{\theta^e}$ (DFH)	0.2087	0.2301	0.1150
CES: $\hat{f}_t^\theta$ vs $\hat{q}_t^\theta$	0.1665	0.1692	0.1831
<i>Panel B: Exclude Covid (GMM AR(3))</i>			
Data: f_t vs q_t (total hires)	8.0161	10.2015	12.2837
Data: f_t vs q_t^u (UE hires)	0.5869	0.7468	0.8993
CD: $\hat{f}_t^\theta$ vs $\hat{q}_t^\theta$	0.1251	0.1662	0.1779
CES: $\hat{f}_t^\theta$ vs $\hat{q}_t^\theta$	0.0890	0.1393	0.1413

Notes: Each cell reports $(SD_f/SD_q)^2$, HP filter with $\lambda = 100,000$. f_t : job-finding rate; q_t : vacancy-filling rate (total hires, `lnqm`); q_t^u : vacancy-filling rate from unemployment outflows only (`lnqm_hu`). $\hat{f}_t^\theta = \mu + \alpha \ln \theta_t$ (theta-only fitted JFR); $\hat{q}_t^\theta = \hat{f}_t^\theta - \ln \theta_t$ (CRS identity). $\hat{f}_t^{\theta^e} = \log f_t - \log \tilde{r}_t$ (DFH q -index, Full Sample α , GMM AR(4)); $\hat{q}_t^{\theta^e} = \hat{f}_t^{\theta^e} - \ln \theta_t$. JFR measures: EHS (2015) = `lnpue`, CPS-Micro = `lnjfsa`, CPS-Macro = `lnfsa`. Panel A: CD GMM AR(4), CES GMM AR(3); Panel B: CD GMM AR(3), CES GMM AR(2), sample ends 2019Q4.

To assess how well a standard matching function with constant matching efficiency

accounts for this moment, I compute the standard deviation and the variance ratio of the fitted value of JFR and VFR from matching function estimation. The method of constructing the fitted values is the same as in previous exercise. For example, if matching function is Cobb-Douglas, the counterfactual JFR is $\hat{f}_t = \exp(\hat{\mu}_{GMM} + \hat{\alpha}_{GMM} \log \theta_t)$, and the counterfactual VFR is $\hat{q}_t = \hat{f}_t / \theta_t$. I use HP filter to detrend the actual data and the counterfactual series, the results with Band-Pass filter are similar and reported in the Appendix.

The variance ratio of θ -only counterfactual JFR and VFR are reported in the last two rows in each panel of Table ???. Under the Cobb-Douglas specification, the predicted variance ratio is around 0.05 in the full sample, and around 0.12-0.18 in the pre-Covid period. Both are far below its empirical counterpart. Allowing for a CES specification modestly increases the variance ratio to around 0.2, but still falls well short of matching the data. The variance ratio significantly improved after controlling for search intensity but still lower than the observed variance ratio.

3 Model

3.1 Environment

Time is discrete, $t = 0, 1, 2, \dots$. The economy consists of a continuum of infinitely lived, risk-neutral workers and firms who discount future payoffs at a common rate $r > 0$. Workers and firms are homogeneous in match quality, so that all matched firm-worker pairs generate identical output in a given period.

3.1.1 Production Technology

Firms produce with labor as the only input under a homogeneous technology, so the flow output of each matched firm-worker pair equals aggregate labor productivity z_t . Assume that the aggregate productivity z_t follows a time-homogeneous, first-order, finite-state Markov chain with support $Z = \{z_1, \dots, z_N\}$ and transition matrix $\pi_{z, z'}$. States are ordered such that $0 < z_1 < z_2 < \dots < z_N$, and the initial state is drawn from the invariant distribution $\mathcal{F}_z$. I assume monotone transition dynamics: for any $z_i < z_j$, the transition distribution conditional on z_j first-order stochastically dominates (FOSD) that conditional on z_i , i.e.,

$$\sum_{z' \leq z_k} \pi_{z_j, z'} \leq \sum_{z' \leq z_k} \pi_{z_i, z'} \quad \forall z_k \in Z, z_j > z_i.$$

This assumption implies persistence in aggregate productivity: higher current states are associated with stochastically better future prospects.

3.1.2 Search and Matching

At the beginning of each period, a measure u_t of unemployed workers search for jobs, and firms post a measure v_t of vacancies. Contacts between unemployed workers and

firms holding vacancies are governed by a Cobb–Douglas matching technology, and the total number of contacts formed in period t is

$$m(u_t, v_t) = Av_t^\alpha u_t^{1-\alpha},$$

where $\alpha \in (0, 1)$ and $A > 0$. Let labor market tightness be $\theta_t = v_t/u_t$. The probability that an unemployed worker meets a vacancy in period t is

$$f(\theta_t) = \frac{m(u_t, v_t)}{u_t},$$

and the probability that a firm holding a vacancy meets an unemployed worker is

$$q(\theta_t) = \frac{m(u_t, v_t)}{v_t}.$$

A contact leads to a match only if the worker and the firm reach agreement in the subsequent wage bargaining stage. Let $\eta_t \in [0, 1]$ denote the probability of agreement conditional on contact, which will be endogenously determined in the bargaining game. The effective job-finding and vacancy-filling probabilities are therefore $f_t = \eta_t f(\theta_t)$ and $q_t = \eta_t q(\theta_t)$.

If agreement is reached, production takes place immediately. Otherwise, the worker remains unemployed, receives a constant flow payoff $b \in (0, z_1)$, and continues searching in the next period. Firms incur a per-period cost $\kappa > 0$ to post and maintain a vacancy. If no agreement is reached, the vacancy remains unfilled and the firm continues searching in the next period subject to the posting cost. An existing match is destroyed at the end of each period with exogenous and constant probability $\sigma > 0$. Upon separation, the worker becomes unemployed and resumes search in the following period. The sequence of events within each period is illustrated in Figure 5.

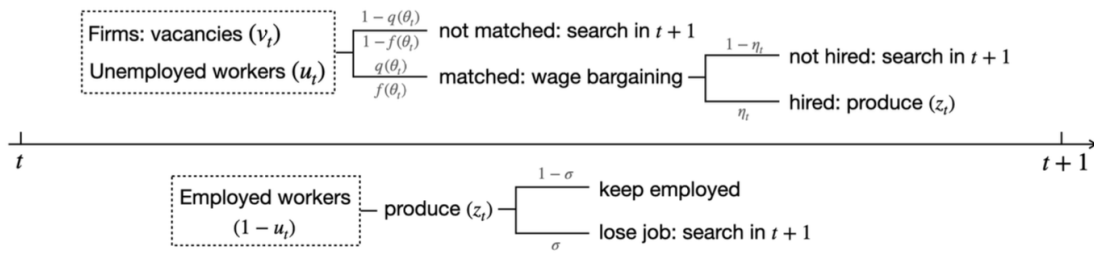


Figure 5: Timeline of economy

3.2 Value Functions under Complete Information

3.2.1 Firm's Value Functions

Consider a match formed in period t . The present value of expected output generated by this match is

$$P_t = z_t + \mathbb{E}_t \left[\sum_{k=1}^{\infty} \frac{(1-\sigma)^k}{(1+r)^k} z_{t+k} \right],$$

where z_s denotes aggregate labor productivity in period s , $\sigma \in (0, 1)$ is the exogenous separation probability, and $\mathbb{E}_t$ denotes rational expectations conditional on information at time t .

Since aggregate productivity follows a first-order Markov chain, the value function depends only on the current state. Hence we can write $P_t = P(z_t)$, where $P(z)$ satisfies the following recursive equation

$$P(z) = z + \frac{1-\sigma}{1+r} \sum_{z' \in Z} \pi_{z,z'} P(z'). \quad (7)$$

Standard contraction arguments imply existence and uniqueness of $P(z)$; see Appendix B for details. Under the monotone transition (FOSD) assumption, $P(z)$ is strictly increasing in z .

Let w_t denote the flow wage paid in period t . The lifetime wage associated with a newly formed match is defined as the present value of all future flow payments:

$$W_t = w_t + \mathbb{E}_t \left[\sum_{k=1}^{\infty} \frac{(1-\sigma)^k}{(1+r)^k} w_{t+k} \right]. \quad (8)$$

The value of a filled job for the firm is therefore

$$J_t = P(z_t) - W_t.$$

The firm incurs the per-period cost $\kappa > 0$ while the vacancy remains open, and obtains J_t upon a successful hire. So the value of holding a vacancy satisfies

$$V_t = -\kappa + q_t J_t + \frac{1-q_t}{1+r} \mathbb{E}_t \left[\sum_{k=1}^{\infty} \frac{\prod_{n=1}^{k-1} (1-q_{t+n})}{(1+r)^k} \left(-\kappa + q_{t+k} J_{t+k} \right) \right]$$

where $q_t = \eta_t q(\theta_t)$ is the effective vacancy-filling rate at period t . And the value function V_t can be rewritten as the following recursive form (see Appendix B for details)

$$V_t = -\kappa + q_t J_t + \frac{1-q_t}{1+r} \mathbb{E}_t V_{t+1} \quad (9)$$

Thus, W_t summarizes the entire sequence of future wage payments into a single present value, which is the only object relevant for the firm's vacancy creation decision. Any wage path $\{w_{t+k}\}_{k=0}^{\infty}$ that yields the same W_t is ex-ante equivalent from the firm's perspective. This observation allows the bargaining problem to be formulated directly in terms of W_t , rather than the full sequence of flow wages.

3.2.2 Worker's Value Functions

Upon accepting a job, the worker receives the flow wage w_t each period until the match is destroyed. If separation occurs, the worker becomes unemployed and searches again in the next period.

The worker's value function of employment at period t is therefore

$$E_t = w_t + \mathbb{E}_t \left[\sum_{k=1}^{\infty} \frac{(1-\sigma)^{k-1}}{(1+r)^k} ((1-\sigma)w_{t+k} + \sigma U_{t+k}) \right],$$

where $\sigma \in (0, 1)$ is the exogenous separation rate and U_{t+k} denotes the value of unemployment at time $t+k$.

The worker's value function of unemployment satisfies

$$U_t = b + \mathbb{E}_t \left[\sum_{k=1}^{\infty} \frac{\prod_{n=1}^{k-1} (1-f_{t+n})}{(1+r)^k} ((1-f_{t+k})b + f_{t+k}E_{t+k}) \right],$$

where $b \in (0, z_1)$ is the constant unemployment benefit and $f_{t+n} = \eta_{t+n}f(\theta_{t+n})$ denotes the effective job-finding rate at period $t+n$.

The value of employment can be decomposed into two components: the lifetime wage W_t and the continuation value associated with future unemployment risk,

$$E_t = W_t + W_{p,t}, \quad (10)$$

where

$$W_{p,t} = \mathbb{E}_t \left[\sum_{k=1}^{\infty} \frac{\sigma(1-\sigma)^{k-1}}{(1+r)^k} U_{t+k} \right].$$

Both U_t and $W_{p,t}$ admit recursive representations (see Appendix B for details):

$$U_t = b + \frac{1}{1+r} \mathbb{E}_t [f_{t+1}(W_{t+1} + W_{p,t+1}) + (1-f_{t+1})U_{t+1}], \quad (11)$$

$$W_{p,t} = \frac{1}{1+r} \mathbb{E}_t [\sigma U_{t+1} + (1-\sigma)W_{p,t+1}]. \quad (12)$$

Under the Markov structure of aggregate productivity, if the lifetime wage W_t , market tightness θ_t , and matching efficiency η_t depend only on the current state z_t in equilibrium, then all value functions (J_t, V_t, E_t, U_t) can be written as time-invariant functions of z_t . In this case, equilibrium is characterized by a system of recursive equations, which together define a fixed point in the space of state-dependent value functions.

3.2.3 Total Net Surplus of a Match

The total net surplus generated by a match at time t is defined as the joint gain of the firm and the worker relative to their respective outside options:

$$S_t = J_t - V_t + E_t - U_t = P_t - V_t + W_{p,t} - U_t. \quad (13)$$

Importantly, the lifetime wage W_t of the current match cancels out from the surplus expression. The surplus therefore depends only on productivity P_t , the value of holding a vacancy V_t , and the worker's net continuation value $W_{p,t} - U_t$, which reflects the value of future matches. In particular, although the wage of the current match does not affect total surplus, expectations about future equilibrium wages influence S_t through the worker's outside option U_t . The surplus S_t will be divided between the worker and the firm in the wage bargaining stage described below.

3.3 Wage Bargaining

Wage determination takes place after an unemployed worker and a firm make contact in the matching market. The bargaining protocol is a static Nash demand game with incomplete information. Assume that each contacted worker–firm pair bargains independently, with no coordination across pairs within or across periods. This assumption is standard in the search and matching literature; see, e.g., Hall and Milgrom (2008).

3.3.1 Setup and Actions

Consider a worker–firm pair that makes contact in period t . If they reach agreement on the wage, the contact turns into a match and generates a total net surplus S_t . This surplus is defined in equation (13) as the joint gain from forming the match relative to the outside options under full-information rational expectations, taking as given the equilibrium sequence of wages, market tightness, and matching efficiency $\{\theta_{t+k}, W_{t+1+k}, \eta_{t+1+k}\}_{k=0}^{\infty}$.

The worker and the firm simultaneously choose desired surplus shares $(x_t^w, x_t^f) \in [0, 1]^2$ of the net surplus S_t . These shares map one-to-one into a wage demand and offer over the lifetime wage W_t . Specifically, a worker who demands share x_t^w induces a lifetime wage $W_t^d(x_t^w)$ such that

$$W_t^d(x_t^w) + W_{p,t} - U_t = x_t^w S_t, \quad (14)$$

which implies

$$W_t^d(x_t^w) = x_t^w (P_t - V_t) + (1 - x_t^w)(U_t - W_{p,t}).$$

Similarly, a firm that demands share x_t^f induces a lifetime wage $W_t^o(x_t^f)$

$$P_t - W_t^o(x_t^f) - V_t = x_t^f S_t,$$

which implies

$$W_t^o(x_t^f) = (1 - x_t^f)(P_t - V_t) + x_t^f (U_t - W_{p,t}).$$

Demands are compatible if $x_t^w + x_t^f \leq 1$ (equivalently, $W_t^d \leq W_t^o$). In that case, agreement is reached and the lifetime wage is set at the midpoint of the proposed range,

$$W_t = \frac{W_t^d + W_t^o}{2}.$$

If $x_t^w + x_t^f > 1$ (equivalently, $W_t^d > W_t^o$), bargaining breaks down: the contact does not result in a match, both parties obtain their outside options, and they return to search in the next period.

3.3.2 Information Structure

The bargaining game features asymmetric information regarding the worker's evaluation of payoffs from alternative wage demands. Firms correctly evaluate all payoffs and continuation values. Workers, however, perceive these payoffs with noise due to limited information or attention at the bargaining stage. This asymmetry is the key departure from the complete-information Nash demand game Nash (1953), and, as shown in Section 3.4, gives rise to endogenous matching efficiency.

Firms. The firm holds rational expectations: given the equilibrium path of lifetime wages, market tightness, and matching efficiency $\{\theta_{t+k}, W_{t+1+k}, \eta_{t+1+k}\}_{k=0}^{\infty}$, the firm correctly computes all value functions as defined in Eq. (7) to (12), and its beliefs about the equilibrium path are consistent with these value functions. In particular, the firm knows the total net surplus S_t from Eq. (13).

Workers. Workers, by contrast, perceive the payoffs from alternative wage demands with noise. The worker understands the structure of the game—including the agreement rule, the surplus-sharing protocol, and the mapping from surplus shares to lifetime wages—but does not precisely assess the net payoff associated with each feasible demand.

Each worker is characterized by an idiosyncratic type $\varepsilon_t(\cdot)$, a random function that assigns an independent expectation error $\varepsilon_t(x)$ to each feasible surplus share $x \in [0, 1]$ ¹³. The type represents the worker's errors in evaluating the action-level payoffs from alternative wage demands relative to the complete information benchmark defined in Eq. (10) and (11).

Nature draws $\varepsilon_t(\cdot)$ independently across workers and over time, and the type is realized before the bargaining game. Upon observing the realized type, the worker chooses a surplus share x_t^w . The realized type is worker's private information, and firms only know the distribution of $\varepsilon_t(\cdot)$.

Specifically, I assume that the expectation error $\varepsilon_t(x)$ has mean zero and follows a Type-I Extreme Value (Gumbel) distribution¹⁴:

¹³The action-contingent structure—one independent error per demand, rather than a single scalar perturbation common to all demands—is a substantive assumption. A scalar perturbation, for instance a noisy signal about the surplus S_t , would induce perfectly correlated errors across demands and generate threshold-rule behavior rather than the choice probabilities that arise under the i.i.d. specification, which naturally results in change in extensive margin of matching efficiency. The action-contingent structure is rationalized by the micro-foundations discussed below; see also the discussion of precedents in the random utility, game theory, and structural labor literatures.

¹⁴The Gumbel distributional assumption is made primarily for analytical tractability, as it yields closed-form choice probabilities. Alternative specifications, such as normally distributed errors, generate similar qualitative predictions.

$$\varepsilon_t(x) \stackrel{i.i.d.}{\sim} \text{Gumbel}\left(0, \frac{1}{\lambda_t^2}\right), \quad (15)$$

where the scale parameter $1/\lambda_t^2$ governs the dispersion of the worker's perceived payoffs. As $\lambda_t \rightarrow \infty$, the model converges to the complete-information benchmark. The scale parameter $1/\lambda_t^2$ is exogenous and may vary over time; its specification and cyclical properties are discussed in Section 3.4.

Two features of this specification deserve emphasis.

(1) *Unbiased expectation.* The zero-mean assumption ensures that the worker's expectations are correct on average. The shock $\varepsilon_t(x)$ introduces dispersion but not systematic bias: uncertainty affects the variance of perceived payoffs, not their mean. As a result, the worker does not systematically over- or under-value any particular wage demand relative to the rational-expectations benchmark.

(2) *Separation of bargaining noise from equilibrium fundamentals.* The expectation error ε_t distorts the worker's perceived payoff *only at the bargaining stage of the current match*. Conditional on the equilibrium path $\{\theta_{t+k}, W_{t+1+k}, \eta_{t+1+k}\}_{k=0}^\infty$, the shock does not enter the value functions $(P_t, V_t, W_{p,t}, U_t)$ or the total net surplus S_t , which are computed under rational expectations as defined in the previous section. Because vacancies are posted at the beginning of each period before bargaining takes place, market tightness θ_t is pre-determined at the bargaining stage. Moreover, each contacted worker-firm pair bargains independently, with no coordination across pairs within or across periods, so the future equilibrium sequence $\{\theta_{t+1+k}, W_{t+1+k}, \eta_{t+1+k}\}_{k=0}^\infty$ can also be taken as primitives for the bargaining game of any individual match at period t . Therefore, the surplus S_t is a *fundamental* object for the current match that serves as the "pie" to be divided. The expectation error affects only *how* an individual worker perceives the payoffs from alternative demands over this pie. However, at equilibrium, ε_t influences the value functions *indirectly* through the equilibrium path of wage W_t , matching efficiency η_t and market tightness θ_t . The full characterization of this feedback is given in Section 3.4.

3.3.3 Payoffs

The payoffs of the bargaining game depend on whether the demands are compatible. Recall from the setup that demands are compatible if $x_t^w + x_t^f \leq 1$, in which case agreement is reached and the lifetime wage is set at the midpoint $W_t = (W_t^d + W_t^o)/2$. If $x_t^w + x_t^f > 1$, bargaining breaks down and both parties obtain their outside options.

Firm's payoff. The firm chooses its surplus share $x_t^f \in [0, 1]$ (equivalently, a wage offer $W_t^o \in \mathcal{W}_t$). Conditional on the worker's demand x_t^w , the firm's payoff is

$$v_f(x_t^f | x_t^w) = \begin{cases} P_t - W_t^m, & \text{if } x_t^w + x_t^f \leq 1, \\ V_t, & \text{if } x_t^w + x_t^f > 1, \end{cases} \quad (16)$$

where $W_t^m = (W_t^d(x_t^w) + W_t^o(x_t^f))/2$ is the agreed lifetime wage and V_t is the value of a vacancy as firm's outside option. The firm's net payoff is the gain from hiring relative to continuing to hold a vacancy. Since the firm has complete information, its payoff is deterministic conditional on the worker's action x_t^w .

Worker's payoff. The worker chooses a surplus share $x_t^w \in [0, 1]$. The worker's true payoff, conditional on the firm's demand x_t^f , is

$$v_w(x_t^w | x_t^f) = \begin{cases} W_t^m + W_{p,t}, & \text{if } x_t^w + x_t^f \leq 1, \\ U_t, & \text{if } x_t^w + x_t^f > 1, \end{cases} \quad (17)$$

where $W_{p,t}$ is the continuation value associated with future unemployment risk and U_t is the value of unemployment. The worker's true payoff is the gain from employment relative to remaining unemployed. As described in the information structure, the worker does not observe v_w directly. Instead, the worker's perceived payoff from demanding share x_t^w is

$$u_w(x_t^w | x_t^f) = v_w(x_t^w | x_t^f) + \varepsilon_t(x_t^w), \quad (18)$$

where $\varepsilon_t(x_t^w)$ is the idiosyncratic expectation error defined in Section 3.3.2. The worker chooses x_t^w to maximize the perceived payoff u_w .

3.3.4 Strategies and Solution Concept

A strategy for the firm is a surplus share $x_t^f \in [0, 1]$. A strategy for the worker is a measurable function $x_t^w : \mathcal{E} \rightarrow [0, 1]$ that maps the realized type $\varepsilon_t(\cdot)$ into a surplus share, where $\mathcal{E}$ denotes the space of type realizations. Upon observing the realized type, the worker chooses the surplus share that maximizes the perceived payoff $u_w(x_t^w | x_t^f)$ defined in Eq. (18). The solution concept is Bayesian Nash Equilibrium (BNE).

Definition 1. The strategy profile $(x_t^{w*}(\cdot), x_t^{f*})$ is a (pure strategy) Bayesian Nash Equilibrium if for all $\varepsilon_t \in \mathcal{E}$,

$$x_t^{w*}(\varepsilon_t) \in \arg \max_{x \in [0, 1]} u_w(x | x_t^{f*}),$$

and

$$x_t^{f*} \in \arg \max_{x \in [0, 1]} \int_{\mathcal{E}} v_f(x | x_t^{w*}(\varepsilon_t)) dF(\varepsilon_t),$$

where F denotes the common prior over the worker's type.

Each worker type plays a pure strategy, choosing the single demand that maximizes perceived payoff given the realized type. From the firm's perspective, however, the worker's action is stochastic because the type is unobserved¹⁵. Under the Gumbel distributional as-

¹⁵The distribution of worker actions induced by pure-strategy BNE in this incomplete-information game is equivalent to the choice probabilities in a Quantal Response Equilibrium (QRE) of the corresponding complete-information game (McKelvey & Palfrey, 1995, 1996). In QRE, a representative agent randomizes over demands according to logit probabilities; the present BNE formulation micro-founds this randomization

sumption on $\varepsilon_t(x)$, the distribution of the worker's demand takes the logit form ¹⁶.

3.3.5 Bayesian Nash Equilibrium

We now characterize the BNE of the bargaining game, taking the total net surplus $S_t > 0$ and the dispersion parameter $\lambda_t > 0$ as given.

Worker's optimal demand. Given the firm's strategy x_t^f , each worker type $\varepsilon_t \in \mathcal{E}$ chooses the surplus share that maximizes the perceived payoff $u_w(x \mid x_t^f)$. Since $\varepsilon_t(x)$ is i.i.d. Gumbel across $x \in [0, 1]$, the standard result from discrete choice theory extends to the continuous case: the distribution of the worker's demand, from the firm's perspective, is given by a logit density,

$$f(x^w \mid x_t^f, \lambda_t) = \frac{\exp(\lambda_t v_w(x^w \mid x_t^f))}{\int_0^1 \exp(\lambda_t v_w(x \mid x_t^f)) dx}, \quad x^w \in [0, 1], \quad (19)$$

where $v_w(x^w \mid x_t^f)$ is the worker's true payoff defined in Eq. (17). Note that v_w is piecewise: it is strictly positive and linear in x^w for $x^w \leq 1 - x_t^f$ (the agreement region) and constant for $x^w > 1 - x_t^f$ (disagreement). The logit density therefore assigns strategy-contingent weight to demands in the agreement region and a constant weight to demands in the disagreement region. The standard discrete choice derivation applies without modification, since the Gumbel distributional assumption requires only measurability of the deterministic payoff component.¹⁷ As the precision λ_t increases, the density concentrates around the worker's best response under complete information; as $\lambda_t \rightarrow 0$, the density approaches the uniform distribution over $[0, 1]$.

Firm's optimal demand. Given the worker's logit demand distribution in Eq. (19), the firm chooses its surplus share to maximize expected profit:

$$x_t^{f*} \in \arg \max_{x^f \in [0, 1]} \int_0^1 v_f(x^f \mid x^w) f(x^w \mid x^f, \lambda_t) dx^w. \quad (20)$$

The firm faces a trade-off: demanding a larger share x_t^f increases the firm's payoff conditional on agreement but raises the probability of disagreement, since agreement requires

as the outcome of heterogeneous types each playing a pure strategy. See Goeree, Holt, and Palfrey (2016) for a comprehensive treatment.

¹⁶Under the alternative assumption of normally distributed errors, the distribution of worker demands would take the probit. The choice between Normal and Gumbel errors is less consequential under the BNE formulation than under QRE, as the representative agent is typically interpreted as the average agent in the economy. The Normal distribution is closed under averaging, however, the Gumbel distribution is closed under the maximum but not under averaging, i.e., the maximum of i.i.d. Gumbel random variables is Gumbel. Adopting a Gumbel specification in a QRE framework therefore implicitly selects the most extreme rather than the average agent as representative.

¹⁷See, e.g., McFadden (1974) and Train (2009). The piecewise structure of v_w introduces a kink at $x^w = 1 - x_t^f$ but does not affect the derivation of the logit density, which requires only that the deterministic component be Borel measurable.

$x_t^w + x_t^f \leq 1$. The optimal demand balances these margins, and the firm's optimization problem is strictly convex which establish the existence and uniqueness of BNE.

The following theorem establishes the main properties of the BNE.

Theorem 1. *For any $\lambda_t > 0$ and $S_t > 0$, there exists a unique pure-strategy BNE $(x_t^{w*}(\cdot), x_t^{f*})$. Define the matching efficiency*

$$\eta(\lambda_t, S_t) = \Pr(x_t^w + x_t^{f*} \leq 1) = \int_0^{1-x_t^{f*}} f(x^w | x_t^{f*}, \lambda_t) dx^w,$$

where $f(x^w | x_t^{f*}, \lambda_t)$ is defined in Eq (19) and the firm's expected bargaining power

$$\beta(\lambda_t, S_t) = \mathbb{E} \left[\frac{x_t^{f*} + (1 - x_t^w)}{2} \mid x_t^w + x_t^{f*} \leq 1 \right].$$

Then:

$$\frac{\partial \eta(\lambda_t, S_t)}{\partial \lambda_t} > 0, \quad \frac{\partial \beta(\lambda_t, S_t)}{\partial \lambda_t} > 0, \quad (21)$$

$$\frac{\partial \eta(\lambda_t, S_t)}{\partial S_t} > 0, \quad \frac{\partial \beta(\lambda_t, S_t)}{\partial S_t} > 0. \quad (22)$$

Proof. See Appendix. □

In what follows, I will write $\eta_t \equiv \eta(\lambda_t, S_t)$ and $\beta_t \equiv \beta(\lambda_t, S_t)$ to emphasize the time-varying nature of matching efficiency and bargaining power in general equilibrium.

Equilibrium lifetime wage. The bargaining outcome pins down the equilibrium lifetime wage. Since the firm's effective share of the surplus is β_t , the firm's value of a filled job is

$$J_t = \beta_t S_t + V_t, \quad (23)$$

and the worker's net gain from employment is

$$E_t - U_t = (1 - \beta_t) S_t. \quad (24)$$

Substituting $J_t = P_t - W_t$ and rearranging yields the equilibrium lifetime wage:

$$W_t = (1 - \beta_t)(P_t - V_t) + \beta_t(U_t - W_{p,t}). \quad (25)$$

This expression is the analogue of the Nash bargaining wage equation, with the key difference that the bargaining share β_t is endogenous and state-dependent rather than a fixed parameter. The lifetime wage is a weighted average of the worker's contribution to the match net of the firm's outside option, $P_t - V_t$, and the worker's outside option net of continuation value, $U_t - W_{p,t}$, with the weight determined by the firm's bargaining power β_t .

The comparative statics (η_t, β_t) in Theorem 1 are the key economic results of the bargaining game. Higher precision λ_t concentrates the worker's demand distribution around the complete-information best response, reducing the probability of incompatible demands and increasing the matching probability η_t . At the same time, as the worker's demands become more predictable, the firm can extract a larger expected share of the surplus, so β_t rises.

The effect of the total surplus S_t operates through the same channel. A larger surplus raises the worker's payoff from agreement relative to disagreement, shifting the logit density toward more moderate demands and expanding the region of compatibility. This directly increases η_t . The same shift also improves the firm's trade-off: with a lower disagreement risk at any given demand, the firm optimally raises its own claim x_t^{f*} . The increase in the firm's demand, combined with the more moderate distribution of worker demands conditional on agreement, jointly push β_t upward. Indeed, in this setup η_t and β_t always move in the same direction: any force that expands the agreement region simultaneously allows the firm to extract a larger share of the surplus.

Figures 6 and 7 illustrate these effects. Figure 6 plots the worker's equilibrium demand density for several values of λ_t , normalizing $S_t = 1$. As λ_t increases, the density sharpens around the worker's best response. Figure 7 plots the firm's expected payoff as a function of its demand x_t^f for the same values of λ_t , showing that the optimal demand and the level of expected profit both increase with precision. The corresponding figures with fixed λ_t and varying S_t are qualitatively similar, as the two operate through the same channel.

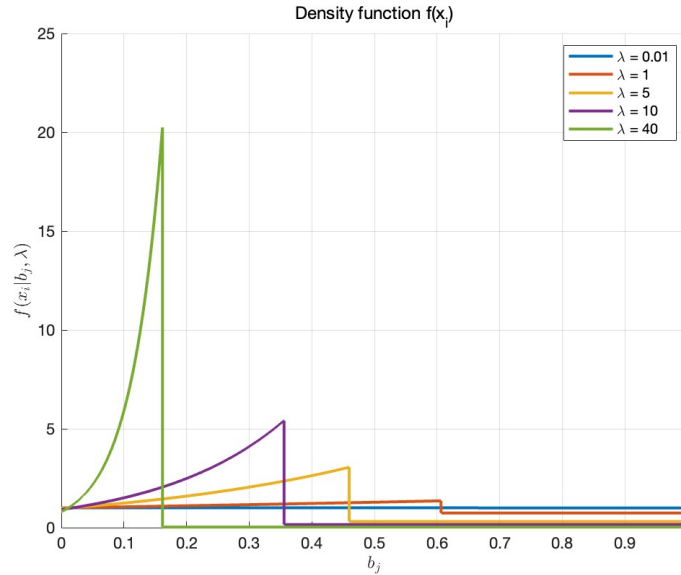


Figure 6: Worker's equilibrium demand density for different values of λ_t , with $S_t = 1$.

3.3.6 Micro-foundations for ε_t

The expectation error $\varepsilon_t(x)$ is a reduced-form representation of dispersion in worker expectations at the bargaining stage. While the Gumbel specification is adopted for ana-

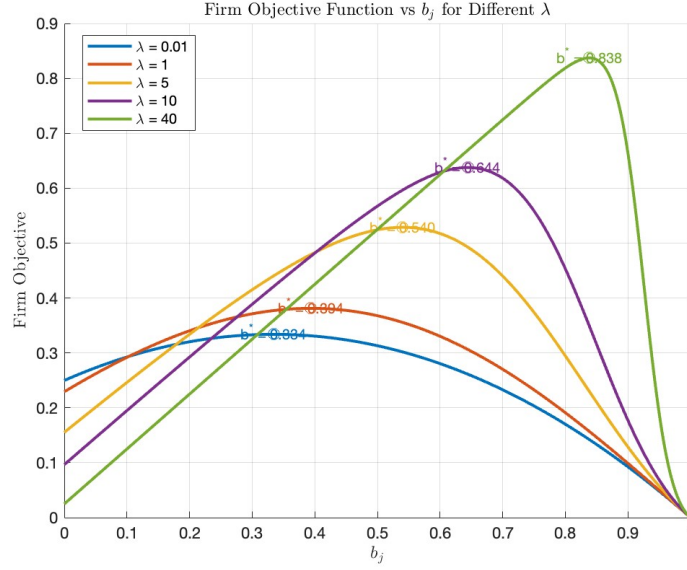


Figure 7: Firm's expected payoff as a function of x_t^f for different values of λ_t , with $S_t = 1$.

lytical tractability, both the additive action-contingent structure and the specific distributional form can be motivated by well-established traditions in the random utility, game theory, and structural labor literatures, as well as by several distinct micro-foundations.

Precedents for the action-contingent structure. The specification of an independent, additively separable payoff perturbation for each available action is a standard modeling device across several literatures. In the random utility framework of McFadden (1974), each alternative carries its own unobserved utility component, and the i.i.d. Gumbel assumption yields the multinomial logit model. This discrete-choice structure is widely adopted in the structural labor literature as a conventional approach to modeling worker choice among discrete alternatives; see, e.g., ? and Todd and Wolpin (2023). In game theory, Harsanyi (1973) introduced games with randomly disturbed payoffs, where each player's payoff to each pure strategy is perturbed by a privately observed, strategy-specific shock; mixed-strategy equilibria of the unperturbed game arise as limits of pure-strategy equilibria of the perturbed game. The structural quantal response equilibrium of McKelvey and Palfrey (1995) builds directly on this additive random-utility foundation: each player observes a randomly perturbed payoff for each available strategy and chooses optimally given these noisy observations. The present specification follows these precedents by assigning each feasible wage demand its own independent payoff perturbation.

(i) *Rational inattention and costly information acquisition.* The primary micro-foundation is that workers allocate limited attention when evaluating alternative wage demands. In the rational inattention framework of Sims (2003), an agent with a finite information-processing capacity optimally acquires a noisy signal about the payoffs of available actions. Matějka and McKay (2015) show that when the agent's information cost is measured by the reduction in Shannon entropy (mutual information) between the prior and

posterior over states, the optimal choice probabilities take the multinomial logit form. In this interpretation, λ_t^2 is proportional to the agent's information-processing capacity: a worker with greater capacity acquires more precise signals and makes choices closer to the full-information optimum. The Gumbel distribution of $\varepsilon_t(x)$ then emerges not as an assumption but as a *consequence* of optimal information acquisition under entropy costs. The worker's optimal posterior, given limited capacity, induces perceived payoffs that equal the true payoff plus an effective noise term that is i.i.d. Gumbel across alternatives, generating the additive perturbation structure in Eq. (18) endogenously. Importantly, this micro-foundation also rationalizes the action-contingent structure: the independence of errors across alternatives is a consequence of optimal attention allocation, not an ad hoc assumption.

(ii) *Wage-contingent preference shocks.* An alternative interpretation is that the worker's valuation of a particular wage demand reflects not only its pecuniary payoff but also non-pecuniary factors that vary across alternatives. These may include the perceived fairness of the proposed wage, psychological reference-point effects, social comparison with peers, or idiosyncratic attitudes toward risk associated with different demand levels. If these non-pecuniary components are additively separable from the monetary payoff and are i.i.d. across alternatives, they generate the payoff perturbation structure in Eq. (18). Under this interpretation, $\varepsilon_t(x)$ captures genuine heterogeneity in worker preferences over wage demands that is orthogonal to economic fundamentals.

(iii) *Noisy signals about economic fundamentals.* Finally, workers may not perfectly observe the current aggregate productivity state z_t or the prevailing equilibrium market conditions (θ_t, η_t) , and instead receive private, noisy signals about these fundamentals at the time of bargaining. Signal noise induces heterogeneity in perceived payoffs across wage demands, which can be approximated by an additive, mean-zero error term under standard distributional assumptions.

These interpretations are not mutually exclusive, and the model is agnostic about which mechanism dominates. What matters for the analysis is the common reduced-form implication: the worker's choice of surplus share is governed by a logit probability, with the precision parameter λ_t controlling the degree of deviation from the full-information best response.

3.4 General Equilibrium

The bargaining game characterized in Section 3.3.5 determines the matching probability η_t and the firm's bargaining power β_t as functions of the total surplus S_t and the precision parameter λ_t . To close the model, two additional conditions are required: a free entry condition that pins down market tightness θ_t , and a law of motion for unemployment u_t .

3.4.1 Free Entry

Firms can freely create vacancies at per-period cost $\kappa > 0$. In equilibrium, entry drives the expected value of a vacancy to zero, i.e., $V_t = 0 \forall t$. Substituting the free entry condition into the recursive expression for the vacancy value yields

$$\kappa = q_t J_t = \eta_t q(\theta_t) \beta_t S_t, \quad (26)$$

where the second equality uses $J_t = \beta_t S_t + V_t = \beta_t S_t$ from Eq. (23) and $q_t = \eta_t q(\theta_t)$. This condition equates the cost of posting a vacancy to the expected return: the probability of contacting a worker, $q(\theta_t)$, times the probability of agreement, η_t , times the firm's share of the surplus, $\beta_t S_t$. Since $q(\theta) = A\theta^{\alpha-1}$ is strictly decreasing in θ and $\eta_t, \beta_t, S_t > 0$, Eq. (26) uniquely determines equilibrium market tightness θ_t as a function of the state.

3.4.2 Law of Motion for Unemployment

The unemployment rate evolves according to

$$u_{t+1} = (1 - u_t)\sigma + u_t(1 - \eta_t f(\theta_t)), \quad (27)$$

where the first term captures inflows from separations of employed workers and the second captures unemployed workers who fail to find a job, either because they do not contact a firm or because bargaining breaks down. The effective job-finding rate $\eta_t f(\theta_t)$ reflects both the contact rate $f(\theta_t)$ and the endogenous matching efficiency η_t .

3.4.3 Recursive Equilibrium

Under the Markov structure of aggregate productivity, assume that $\lambda_t = \lambda(z_t)$, I look for a recursive equilibrium in which all equilibrium objects depend on time only through the current aggregate state z_t and the predetermined unemployment rate u_t .

Definition 2. A recursive equilibrium is a collection of functions

$$\{\theta(z), \eta(z), \beta(z), W(z), S(z), P(z), U(z), W_p(z)\}$$

defined on $Z = \{z_1, \dots, z_N\}$, and a law of motion $u' = \Gamma(z, u)$, such that:

(i) **Value functions.** $P(z)$, $U(z)$, and $W_p(z)$ satisfy the recursive equations

$$P(z) = z + \frac{1 - \sigma}{1 + r} \sum_{z' \in Z} \pi_{z, z'} P(z'), \quad (28)$$

$$U(z) = b + \frac{1}{1 + r} \sum_{z' \in Z} \pi_{z, z'} \left[\eta(z') f(\theta(z')) (W(z') + W_p(z')) + (1 - \eta(z') f(\theta(z'))) U(z') \right], \quad (29)$$

$$W_p(z) = \frac{1}{1 + r} \sum_{z' \in Z} \pi_{z, z'} \left[\sigma U(z') + (1 - \sigma) W_p(z') \right]. \quad (30)$$

- (ii) **Surplus.** $S(z) = P(z) + W_p(z) - U(z)$, where free entry ($V(z) = 0$) has been imposed.
- (iii) **Bargaining.** $\eta(z) = \eta(\lambda(z), S(z))$ and $\beta(z) = \beta(\lambda(z), S(z))$ are determined by the unique BNE of the bargaining game (Theorem 1), where $\lambda(z)$ is the state-contingent precision parameter.
- (iv) **Wage.** $W(z) = (1 - \beta(z))(P(z)) + \beta(z)(U(z) - W_p(z))$, from Eq. (25) with $V(z) = 0$.
- (v) **Free entry.** $\theta(z)$ satisfies

$$\kappa = \eta(z) q(\theta(z)) \beta(z) S(z).$$
- (vi) **Law of motion.** $u' = \Gamma(z, u) = (1 - u)\sigma + u(1 - \eta(z)f(\theta(z)))$.

Since $q(\theta)$, η , β , and S all depend only on z , the free entry condition pins down θ as a function of z alone. The unemployment rate u_t scales the total number of vacancies $v_t = \theta(z_t)u_t$ and the total number of matches, but does not affect per-worker or per-vacancy rates. As a result, the value functions and equilibrium objects can be solved first as functions of z alone, and the law of motion for u_t can then be simulated given the solution.

In addition, the recursive equilibrium is a fixed point in the space of state-contingent functions. Conditions (i)–(v) form a system of functional equations in $\{P, U, W_p, S, \eta, \beta, W, \theta\}$. The function $P(z)$ can be solved independently from Eq. (28), as it depends only on the exogenous productivity process. The remaining functions are jointly determined: the surplus $S(z)$ depends on $U(z)$ and $W_p(z)$, which in turn depend on future wages and matching probabilities, which themselves depend on future surpluses through the bargaining game.

Besides, the precision parameter $\lambda(z)$ is taken as a primitive of the equilibrium definition. In the quantitative analysis, I specify $\lambda(z)$ as a parametric function of the aggregate state, allowing the degree of information friction to vary over the business cycle.

4 Quantitative results

To assess the quantitative performance of this model, I assume the log of labor productivity follows the stationary AR(1) process as in Shimer (2005):

$$\log(z_t) = \rho_z \log(z_{t-1}) + \sigma_z \varepsilon_t, \quad \varepsilon_t \stackrel{i.i.d}{\sim} WN(0, 1)$$

The assumption that λ_t is procyclical—equivalently, that uncertainty at the bargaining stage is counter-cyclical—is motivated by a large body of evidence documenting that macroeconomic uncertainty rises in recessions (Bloom, 2009; Jurado et al., 2015). In Appendix ??, I provide additional supporting evidence from a structural VAR. Using a tri-variate SVAR on the growth rates of consumer and business confidence indices and the Jurado et al. (2015) uncertainty series, I find that consumer confidence responds significantly more negatively to uncertainty shocks than business confidence, consistent with households—who are on the worker side of the bargaining table—being more sensitive to uncertainty fluctuations. Moreover, the cross-sectional dispersion of consumer confidence is positively associated with aggregate uncertainty, suggesting that higher uncertainty disperses household expectations. These patterns support the modeling choice that λ_t , which governs the precision of worker expectations at the bargaining stage, comoves positively with the business cycle.

Specifically, I assume that $\log(\lambda_t)$ comoves perfectly with productivity:

$$\log(\lambda_t) = \mu_\lambda + \sigma_\lambda \log(z_t).$$

Compared with the canonical DMP model used in Shimer (2005), I have only two additional parameters to calibrate: $(\mu_\lambda, \sigma_\lambda)$. Since σ_λ governs the time-series dynamics, it cannot be pinned down by matching average statistics in the data to steady-state statistics in the model. Therefore, in the benchmark calibration, I calibrate σ_λ to match the ratio of the standard deviation of log uncertainty to the standard deviation of log productivity. The uncertainty measure is the 1-month ahead square root of forecast error variance constructed by Jurado et al. (2015), as the model is calibrated at monthly frequency.

In Shimer (2005), the firm’s bargaining power is exogenously set to a constant by the Hosios condition, but it is endogenously determined by the aggregate state in my model. I therefore calibrate μ_λ by matching the average bargaining power of the firm. Multiple empirical studies suggest that the Hosios condition is unlikely to hold in the data, and that the firm’s bargaining power is much higher than the matching elasticity with respect to vacancies, α . Using CPS data, C. J. Flinn (2006) estimate the average bargaining power of workers at 0.424, and C. Flinn and Mabli (2008) obtain a similar estimate of 0.45 using SIPP data. Jarosch, Nimczik, and Sorkin (2024) estimate time-varying firm bargaining power using a long panel of Austrian data from 1998 to 2014, finding values ranging between 0.53 and 0.6. Based on this empirical evidence, I calibrate μ_λ by matching an average firm bargaining power of 0.6. This target is conservative relative to the 0.948 implied by Hagedorn and Manovskii (2008), who target the empirical elasticity of wages with respect

to labor productivity.

All other parameters are calibrated in the same way as in Shimer (2005). Two remarks are in order. First, this model also generates endogenous state-contingent matching efficiency, which affects both the job-finding rate f_t and equilibrium market tightness θ_t . I therefore calibrate the constant term A in the matching function $m(u, v) = Au^{1-\alpha}v^\alpha$ to match the average job-finding rate of 0.45 as in Shimer (2005). Second, the flow cost of posting a vacancy κ is calibrated by matching average market tightness to $\theta = 1$ as in Shimer (2005). The parameter values used in the benchmark calibration are reported in Table 8.

Table 8: Benchmark calibration of discrete-time model, monthly frequency

Moment	Data	Model	Parameter	Value
Quarterly std of productivity	0.015	$\sqrt{\frac{\sigma_z^2 \sum_{j=0}^2 \rho_z^{2j}}{1-\rho_z^6}}$	σ_z	0.0058
Quarterly autocorrelation of productivity	0.878	ρ_z^3	ρ_z	0.958
Quarterly std of uncertainty	0.153	$\sigma_\lambda \sigma_z$	σ_λ	10
Average bargaining power of firm	0.6	$\beta(z, \lambda)$	μ_λ	0.2
Unemployment benefit	$0.4\mathbb{E}(w)$	$0.4\mathbb{E}(w)$	b	0.4
Quarterly discount rate	0.012	$(1+r)^3 - 1$	r	0.004
Monthly separation rate	0.034	σ	σ	0.034
Estimated matching elasticity	0.28	α	α	0.28
Monthly job-finding rate	0.45	$\tilde{f}(\theta, \eta)$	A	0.56
Labor market tightness	1	θ	κ	0.74

For the model-generated statistics, I use the free entry condition and the equilibrium outcome of the bargaining game to find the state-contingent market tightness $\theta(z, \lambda)$, matching efficiency $\eta(z, \lambda)$, and firm bargaining power $\beta(z, \lambda)$, and then simulate the model. Specifically, I select an initial aggregate state from the discrete support of z and use the corresponding steady-state unemployment rate as the initial condition. I then generate a random sequence of aggregate productivity shocks $\{z_t\}$ using the discrete transition probability matrix, and compute the unemployment rate using the law of motion. At the end of each month, I record the aggregate state and the unemployment rate. I discard the first 3,000 “months” of data and use the model to generate 636 data points, corresponding to monthly data over 53 years. I detrend the log of the model-generated data using an HP filter with the smoothing parameter 10^5 . I repeat this procedure 10,000 times, yielding reliable estimates of the standard deviations and correlations of model-generated variables.

Table 10 Panel A reports the results from simulations of the model with labor productivity shocks and uncertainty shocks under the benchmark calibration. For comparison, I also report the results for the canonical DMP model with only labor productivity shocks in Panel B of Table 10. The corresponding U.S. data moments are reported in Table 9.

The model performs remarkably well in capturing the comovement of unemployment and vacancies, which is -0.771 in the data, and -0.779 in the full model. In terms of

Table 9: Volatility and correlations of labor market variables, US data 2001-2025

	u	v	θ	$\tilde{f}$	z
Standard deviation	0.2200	0.1904	0.4080	0.1705	0.0156
Quarterly autocorrelation	0.8198	0.9088	0.8701	0.9058	0.8485
Correlation matrix	u	1.0000	-0.7711	-0.9521	0.5096
	v	—	1.0000	0.9172	-0.1850
	v/u	—	—	1.0000	-0.3934
	$f(\theta)$	—	—	—	1.0000
	z	—	—	—	—

volatility, the model with uncertainty shocks generates a standard deviation of unemployment that is 6.3 times higher than that in the baseline model with only labor productivity. This highlights the significant amplification that the information friction generates for labor market fluctuations.

To decompose the sources of amplification, I compute the fraction of unemployment volatility attributable to procyclical matching efficiency η_t versus procyclical bargaining power β_t . In the benchmark calibration, 64.5% of the amplification in unemployment volatility is driven by fluctuations in η_t , with the remainder attributable to β_t . This confirms the mechanism discussed in Section 5: while procyclical β_t amplifies market tightness and helps restore the negative vacancy-unemployment correlation, the dominant source of unemployment amplification is the direct effect of η_t on the job-finding rate.

Beyond volatility, the model also improves along other dimensions relative to the baseline DMP model. The correlation between unemployment and vacancies is substantially closer to its empirical counterpart, reflecting the role of procyclical β_t in generating sufficiently large movements in market tightness and captures well the observed slope of empirical Beveridge curve. In addition, the model produces a more realistic ratio of vacancy volatility to unemployment volatility, as the two channels— η_t acting primarily on unemployment and β_t acting primarily on vacancies—generate differentiated responses across the two variables.

I also consider a counterfactual calibration in which σ_λ is chosen to match the observed standard deviation of market tightness θ_t . Under this calibration, the model accounts for approximately 80% of the observed unemployment volatility in the data, while maintaining a realistic vacancy-unemployment correlation and variance ratio. This is close to the fraction of unemployment volatility attributable to job-finding rates, with the remaining 20% explained by inflows into unemployment (Shimer, 2012). This counterfactual exercise illustrates that the model's amplification mechanism is powerful enough to explain the bulk of unemployment fluctuations once the volatility of market tightness is disciplined by the data.

Table 10: Volatility and correlations of labor market variables, calibration

	u	v	θ	$\tilde{f}$	z
Panel A: Labor productivity and Uncertainty shock					
Standard deviation	0.0341	0.0404	0.0704	0.0380	0.0139
Quarterly autocorrelation	0.8746	0.6687	0.8280	0.8281	0.8278
Correlation matrix	u	1.0000	-0.7791	-0.9324	-0.9316
	v	-	1.0000	0.9529	0.9520
	θ	-	-	1.0000	0.9991
	$\tilde{f}$	-	-	-	1.0000
	z	-	-	-	-
Panel B: Only labor productivity shock					
Standard deviation	0.0054	0.0164	0.0213	0.0060	0.0139
Quarterly autocorrelation	0.8757	0.7788	0.8293	0.8293	0.8281
Correlation matrix	u	1.0000	-0.8845	-0.9330	-0.9323
	v	-	1.0000	0.9931	0.9931
	θ	-	-	1.0000	1.000
	$\tilde{f}$	-	-	-	1.0000
	z	-	-	-	-

Table 11: Key statistics of labor market variables, calibration with θ_t volatility

	u	v	θ	$\tilde{f}$	z
Standard deviation	0.1950	0.1796	0.4080	0.1693	0.0139
Quarterly autocorrelation	0.8657	0.6613	0.8196	0.8264	0.8287
Correlation matrix	u	1.0000	-0.7620	-0.9348	-0.9380
	v	-	1.0000	0.9523	0.9410
	θ	-	-	1.0000	0.9951
	$\tilde{f}$	-	-	-	1.0000
	z	-	-	-	-

5 Mechanism

This section provides an analytical discussion of the amplification mechanism. The equilibrium characterized in Section 3.4 determines matching efficiency η_t and the firm's bargaining power β_t as endogenous, procyclical functions of the aggregate state. I begin by showing how the procyclicality of these objects amplifies labor market fluctuations through the two key equilibrium conditions (Section 5.1). I then relate the model to the Nash bargaining benchmark, in which both objects are constant (Section 5.2). Section 5.3 develops a steady-state elasticity analysis that connects the amplification mechanism to the fundamental surplus framework of Ljungqvist and Sargent (2017), highlighting a new channel that operates outside the traditional surplus-based logic. Section 5.4 discusses how the model addresses two prominent critiques in the literature.

5.1 Amplification through Procyclical η_t and β_t

The dynamics of market tightness and unemployment in the model are governed by two conditions:

$$\kappa = q(\theta_t) \cdot \eta_t \cdot \beta_t \cdot S_t, \quad (\text{FE})$$

$$u_{t+1} = (1 - f(\theta_t) \cdot \eta_t) u_t + \sigma(1 - u_t). \quad (\text{LOM})$$

The free entry condition (FE) equates the cost of posting a vacancy to the expected return: the probability of contacting a worker, $q(\theta_t)$, times the probability of agreement, η_t , times the firm's share of the surplus, $\beta_t S_t$. The law of motion (LOM) governs the evolution of the unemployment rate, where the effective job-finding rate $f(\theta_t) \cdot \eta_t$ reflects both the contact rate and the endogenous matching efficiency.

Both η_t and β_t are procyclical—rising in booms and falling in recessions—as established in Theorem 1. Their procyclicality generates amplification through complementary channels.

The β_t channel: amplifying market tightness. In recessions, lower aggregate productivity reduces the total surplus S_t , and heightened uncertainty (lower λ_t) further reduces the firm's surplus share β_t . Both forces compress the right-hand side of (FE), causing firms to post fewer vacancies and market tightness θ_t to fall sharply. Importantly, firms would like to cut wages aggressively in recessions to preserve hiring incentives, but workers' increased uncertainty limits the scope for wage cuts: noisy wage demands raise the probability of bargaining breakdown, constraining how much of the surplus firms can extract. This endogenous wage rigidity depresses the expected gain from posting a vacancy, amplifying the decline in θ_t beyond what productivity movements alone would generate.

The η_t channel: a direct effect on unemployment. Matching efficiency η_t enters the law of motion (LOM) directly and linearly. In recessions, lower η_t reduces the actual job-finding rate $f(\theta_t) \cdot \eta_t$, causing unemployment to rise even conditional on θ_t . This is a fundamentally different channel from the β_t channel: it operates on the extensive margin of match formation, not through vacancy creation. Because η_t enters (LOM) linearly while θ_t enters through the concave function $f(\theta_t) = A\theta_t^\alpha$, a given percentage decline in η_t has a larger effect on unemployment than an equivalent decline in θ_t .

The combined effect. Conversely, during expansions, higher λ_t and S_t raise both η_t and β_t . Matches form more efficiently, firms capture a larger share of the surplus, and vacancy creation surges. The result is a larger decline in unemployment than a model with constant η and β would produce. This dual mechanism—amplified θ_t fluctuations from procyclical β_t , and direct unemployment effects from procyclical η_t —is the core of the model's ability to generate realistic labor market volatility.

5.2 Link to Nash Bargaining

The standard Nash bargaining solution provides a useful benchmark. When the worker and firm divide the surplus by maximizing the Nash product with exogenous and constant bargaining weights, two restrictions obtain: the firm's bargaining power β is constant across states, and matching efficiency equals one—any contacted pair with positive surplus immediately forms a match. The equilibrium conditions reduce to

$$\begin{aligned}\kappa &= q(\theta_t) \cdot \beta \cdot S_t, \\ u_{t+1} &= (1 - f(\theta_t)) u_t + \sigma(1 - u_t).\end{aligned}$$

Under this specification, the expected profit βS_t is the sole source of fluctuations in market tightness, and market tightness is the sole driver of unemployment dynamics. There is a high “pass-through” from the matching function elasticity α : productivity shocks affect θ_t through the surplus, and θ_t affects u_t through $f(\theta_t) = A\theta_t^\alpha$. This is the common feature of existing solutions to Shimer's puzzle: the model needs large surplus fluctuations combined with a large matching efficiency to generate realistic unemployment volatility.

One might attempt to restore amplification by introducing exogenous matching efficiency shocks η_t . This helps: η_t enters (LOM) directly and amplifies unemployment fluctuations through a channel separate from θ_t . However, exogenous η_t shocks alone generate a *positive* correlation between vacancies and unemployment—the opposite of the observed Beveridge curve pattern. A negative η_t shock raises unemployment (directly through the law of motion) while simultaneously reducing θ_t (through the free entry condition), but because the decline in θ_t is modest, vacancies $v_t = \theta_t u_t$ move in the same direction as u_t . This problem is formalized in Section 5.3.

Restoring the correct sign of the v - u correlation requires that η_t co-moves with $\beta_t S_t$: fluctuations in the expected profit must be sufficiently large to amplify θ_t and dominate the unemployment effect. Under complete information, the Coase theorem implies efficient bargaining whenever the surplus is positive, so $\eta_t = 1$ is hard to relax. In my model, incomplete information breaks this logic: the agreement probability depends on both the surplus and the precision of worker expectations, providing a micro-foundation for procyclical η_t that naturally co-moves with the firm's expected profit.

5.3 Steady-State Elasticities and the Fundamental Surplus

To formalize the amplification mechanism, I analyze comparative statics of the steady-state equilibrium. The steady-state analysis provides transparent closed-form expressions that, as shown by Ljungqvist and Sargent (2017), closely approximate the stochastic dynamics of calibrated DMP models.¹⁸

¹⁸The derivations of all steady-state elasticities stated in this section are collected in Appendix ??.

5.3.1 The Ljungqvist–Sargent Decomposition

Ljungqvist and Sargent (2017) show that in a broad class of matching models, the elasticity of market tightness with respect to productivity admits a two-factor multiplicative decomposition:

$$\varepsilon_{\theta,z} = \Upsilon \cdot \frac{z}{z-x}, \quad (31)$$

where Υ is a factor bounded above by $1/(1-\alpha)$ and x denotes a “fundamental surplus deduction”—the part of output that the invisible hand *cannot* allocate to vacancy creation. All existing solutions to the Shimer puzzle operate by making the fundamental surplus fraction $(z-x)/z$ small: a high value of leisure (?), a sticky wage (Hall, 2005), alternating-offer bargaining (Hall & Milgrom, 2008), or fixed matching costs (?). Despite their apparent diversity, these mechanisms share a single intermediating channel: they amplify the elasticity of θ with respect to productivity by shrinking what is left over for vacancy creation.

My model introduces an amplification channel that operates *outside* the fundamental surplus framework. The key difference is that matching efficiency η_t enters both the free entry condition and the law of motion for unemployment, so the elasticity of u to aggregate shocks is no longer mediated solely through θ .

5.3.2 Unemployment Elasticities: η versus θ

In steady state, the unemployment rate satisfies $u = \sigma/(\sigma + f(\theta)\eta)$. The elasticities of unemployment with respect to market tightness and matching efficiency are

$$\varepsilon_{u,\theta} = -\alpha(1-u), \quad (32)$$

$$\varepsilon_{u,\eta}^{\text{direct}} = -(1-u). \quad (33)$$

The ratio is $1/\alpha$: with $\alpha = 0.5$, a one-percent change in matching efficiency has twice the direct impact on unemployment as a one-percent change in market tightness. In the fundamental surplus framework, the only way to amplify unemployment fluctuations is to increase $\varepsilon_{\theta,z}$ —by making the surplus fraction small. My model breaks this constraint: even if $\varepsilon_{\theta,z}$ is moderate, procyclical η_t provides an independent channel that does not require the surplus fraction to be small.

5.3.3 Equilibrium Response of θ to η and β

The matching efficiency and bargaining power shocks also affect unemployment indirectly through their equilibrium effect on θ . Under the Nash bargaining benchmark with constant β , the steady-state free entry condition implies the following elasticities (see Ap-

pendix ??):

$$\varepsilon_{\theta,\eta} = \frac{r + \sigma}{(r + \sigma)(1 - \alpha) + f(\theta)\eta(1 - \beta)}, \quad (34)$$

$$\varepsilon_{\theta,\beta} = \frac{r + \sigma + f(\theta)\eta}{(r + \sigma)(1 - \alpha) + f(\theta)\eta(1 - \beta)}. \quad (35)$$

Several properties are worth noting. First, $\varepsilon_{\theta,\beta} > \varepsilon_{\theta,\eta}$, since the numerator of (35) exceeds that of (34) by $f(\theta)\eta > 0$. Intuitively, a β -shock shifts the surplus division toward the firm without altering the agreement probability, so it has a pure effect on vacancy creation incentives. An η -shock, by contrast, also enters the surplus equation through the worker's outside option.

Second, $\varepsilon_{\theta,\eta} < 1/(1 - \alpha)$, which limits the indirect effect of η on unemployment. This is precisely why η -shocks alone generate a positive v - u correlation: the decline in θ from a negative η -shock is too small to offset the direct rise in u .

Third, $\varepsilon_{\theta,\beta} > 1$, so β -shocks always generate a negative v - u correlation:

$$|\varepsilon_{u,\beta}| = \alpha(1 - u) \cdot \varepsilon_{\theta,\beta} < \varepsilon_{\theta,\beta},$$

which means that a β -shock moves vacancies by more than unemployment (in percentage terms), ensuring v and u move in opposite directions.

5.3.4 The Combined Elasticity and the Beveridge Curve

Combining direct and indirect effects, the total elasticity of unemployment along the η -shock is

$$\varepsilon_{u,\eta}^{\text{total}} = -(1 - u) - \alpha(1 - u) \cdot \varepsilon_{\theta,\eta} = -\left(1 + \frac{f(\theta)\eta(1 - \beta)}{r + \sigma}\right)(1 - u). \quad (36)$$

With constant matching efficiency, the slope of the model-implied Beveridge curve is $-\alpha(1 - u)$. The additional term $f(\theta)\eta(1 - \beta)/(r + \sigma)$ arises entirely from the equilibrium response to η fluctuations. Since the monthly job-finding rate $f(\theta)\eta$ is an order of magnitude larger than $r + \sigma$ in U.S. data, this term dominates, and the equilibrium elasticity exceeds $\alpha(1 - u)$ by a factor of at least four at standard calibrations.

However, the elasticity in (36) reflects only the η channel. When both η_t and β_t are procyclical, the effective Beveridge curve slope reflects both channels. Because β_t -shocks have a strong effect on θ_t but only a modest indirect effect on u_t , they tilt the equilibrium locus back toward a negative slope. The quantitative section confirms that the combination of both channels—amplified u_t fluctuations from η_t and amplified θ_t fluctuations from β_t —is necessary and sufficient to match the observed Beveridge curve.

5.4 Discussion of Critiques

5.4.1 The CRK Critique

Chodorow-Reich and Karabarbounis (2016) construct a time series of the flow opportunity cost of employment—comprising forgone public benefits and the consumption value of non-working time—and establish two findings that constrain the class of admissible amplification mechanisms. First, the opportunity cost is *procyclical*: it rises in booms and falls in recessions, with an elasticity with respect to labor productivity that is large and positive. Second, and more importantly, this procyclicality implies that the net match surplus $S_t = P_t + W_{p,t} - U_t$ fluctuates by substantially less than productivity itself, because the worker's outside option moves in the same direction as output. Their central conclusion is that models which generate unemployment volatility primarily by making the surplus large and volatile—such as ?, who calibrate the opportunity cost close to productivity to shrink the fundamental surplus—are inconsistent with the measured cyclicity of the opportunity cost. The critique applies broadly: *irrespective of the level* of the opportunity cost, its estimated procyclicality implies far less unemployment volatility in leading search models than observed in the data.

My model is consistent with the CRK critique for two reasons. First, the primary source of amplification in the quantitative analysis is procyclical matching efficiency η_t , driven by counter-cyclical uncertainty λ_t . This channel operates independently of the surplus level or its cyclicity: even if S_t fluctuates modestly (as the CRK findings imply), η_t provides a direct, additional source of variation in the job-finding rate that does not require the surplus to be large or volatile. Second, procyclical β_t amplifies market tightness without requiring a small fundamental surplus fraction. The β_t channel operates through the *division* of the surplus, not its *size*: a given surplus generates more vacancy creation in booms (when β_t is high) and less in recessions (when β_t is low). This is distinct from the mechanisms catalogued in Ljungqvist and Sargent (2017), all of which amplify by shrinking the fundamental surplus.

Two caveats are in order. The CRK analysis is conducted in a two-state framework with Nash bargaining, and the mapping to my model—which features endogenous matching efficiency and state-dependent bargaining power—is not one-to-one. In particular, the object $U_t - W_{p,t}$ in my model includes the endogenous matching probability η_t in the worker's continuation value, which the CRK measurement does not account for. Nevertheless, the qualitative insight carries over: the model does not rely on a small or acyclical opportunity cost to generate amplification.

5.4.2 The Wage Rigidity Critique

A prominent class of solutions to the Shimer puzzle introduces real wage rigidity to prevent wages from absorbing productivity shocks, thereby amplifying fluctuations in the firm's expected profit and hence in vacancy creation. Hall (2005) proposes a fixed wage norm, Hall and Milgrom (2008) introduce alternating-offer bargaining with credible

threats, and subsequent work incorporates staggered wage contracts and related mechanisms. ? challenges this approach on empirical grounds: microeconomic evidence shows that wages in new matches are procyclical and volatile, consistent with flexible bargaining but inconsistent with the degree of rigidity needed to resolve the puzzle. His central argument is that explanations of unemployment volatility must preserve the observed cyclicalities of wages in new matches.

However, the empirical basis for this critique has been questioned. Gertler et al. (2020) show that the apparent high cyclicalities of new hires' wages is driven by composition bias: in booms, a larger share of new hires are job-to-job changers who move into better matches, mechanically raising the average new-hire wage. Once this composition effect is controlled for—by isolating new hires from unemployment, who provide a composition-free estimate of wage flexibility—wages of new hires are no more cyclical than those of incumbent workers. This suggests that wage rigidity at the relevant margin may be more empirically plausible than ? argued. Regardless of where this debate settles, my model does not hinge on it: the primary source of amplification is procyclical matching efficiency η_t , a channel that is entirely orthogonal to the degree of wage rigidity.

My model addresses this critique by generating amplification through two channels, only one of which involves wage rigidity. The endogenous wage rigidity from procyclical β_t —wages respond less to productivity than under constant bargaining power—is a by-product of the information friction, not an imposed assumption. As shown in the quantitative analysis, the major source of amplification is procyclical matching efficiency η_t , which operates on the extensive margin of match formation and does not require rigid wages. In the model, the equilibrium wage $W_t = (1 - \beta_t)P_t + \beta_t(U_t - W_{p,t})$ is *flexible*: it varies with the aggregate state through both P_t and β_t . The degree of wage rigidity is determined endogenously by the precision parameter λ_t , and the model does not require implausibly rigid wages to match the data. The amplification comes primarily from the η_t channel—a mechanism that is entirely absent in the wage rigidity literature.

6 Conclusions

This paper highlights the critical role of matching efficiency in understanding unemployment volatility, a factor often overlooked in the existing literature. While canonical Diamond-Mortensen-Pissarides (DMP) models have struggled to quantitatively replicate observed unemployment volatility, many papers make attempts to solve the excess volatility puzzle, but they all rely on empirically implausible elasticity of matching to vacancy, and thus all the models overstate the role played by the vacancy creation. I argue that the procyclical matching efficiency are essential to close the gap. My empirical analysis underscores the significant contribution of matching efficiency to overall unemployment volatility, demonstrating that it accounts for a substantial portion of the fluctuations in the job-finding rate. In addition, the empirical evidence also supports the model mechanism with procyclical matching efficiency.

Therefore, I propose a novel theoretical framework based on a modified Nash demand game with incomplete information. This micro-founded approach endogenously generates procyclical matching efficiency and, implicitly, a degree of downward wage rigidity. The core mechanism revolves around counter-cyclical uncertainty about bargaining pay-offs: during economic expansions, reduced uncertainty leads to fewer bargaining mistakes and a higher probability of successful agreement (increased matching efficiency); conversely, during recessions, heightened uncertainty impedes agreement. This allows my model to generate fluctuations in matching efficiency consistent with empirical observations, providing a solution that aligns with both the data and existing critiques regarding the quantitative shortcomings of labor market models.

My findings suggest that by explicitly modeling the micro-level bargaining process under incomplete information, I can capture a richer set of labor market dynamics that are crucial for understanding business cycle fluctuations. This approach not only provides a theoretical foundation for procyclical matching efficiency but also offers a pathway to resolve Shimer's puzzle without resorting to extreme parameter values. As shown in the quantitative results, the model with uncertainty in bargaining outcomes amplifies unemployment volatility by a factor of 6.3 compared to the baseline model, even under the conservative calibration strategy adopted by Shimer (2005). This highlights the quantitative success of this model. If I further match the observed volatility of matching efficiency in the data, the model with uncertainty shock is able to account for more than 60% of observed unemployment volatility in the data.

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